

NATURAL DRAFT FIRED HEATER VER 1

Operator Training Simulator · H-100

Chulalongkorn University · Department of Chemical Engineering

HAZOP REPORT

Hazard and Operability Study of the H-100 fired heater

DOCUMENT NO. : NDFH-V1-SA-HZP-001
DOCUMENT TITLE : HAZOP REPORT
STANDARD : IEC 61882:2016
DRAWINGS : P&ID-001 rev C · NDFH-V1-PID-001 rev D
STUDY TYPE : Teaching study, basic-design level

Teaching document generated from the HAZOP tab of the H-100 simulator. It is not a study of a real plant.

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
A	03-Oct-26	Issued for teaching	A. Soottitantawat			

HAZOP CLOSE-OUT SUMMARY

Project: Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator · Document NDFH-V1-SA-HZP-001 rev A · 27 actions, all open at issue. The action party fills "Action taken" and the HAZOP leader changes the status to CLOSED when the close-out sheet (Appendix 3) is signed.

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1 Feed supply: TK-101 → P-100A/B → E-100 inlet	R01	N1.1	Auto-start spare P-100B on low discharge pressure (PI-002) or P-100A trip, with the start completing inside the FSLL-008 delay; test the spare monthly.	Instrument / Operations		OPEN	Risk 8 Medium
N1 Feed supply: TK-101 → P-100A/B → E-100 inlet	R02	N1.1, N1.2, N1.6, N1.7, N1.8	Add level transmitter LT-101 on TK-101 with low-level alarm LAL and low-low pump trip LSL to protect P-100A/B from running dry.	Instrument		OPEN	Risk 9 Medium
N1 Feed supply: TK-101 → P-100A/B → E-100 inlet	R03	N1.5	Provide minimum-flow recirculation (restriction orifice back to TK-101) on P-100A/B so a closed FV-010 cannot deadhead the running pump.	Process		OPEN	Risk 6 Medium
N1 Feed supply: TK-101 → P-100A/B → E-100 inlet	R04	N1.4, N3.4	Confirm a check valve on each pump discharge; include in the inspection plan (passing check valve spins the idle pump backwards).	Mechanical		OPEN	Risk 4 Low
N2 E-100 feed preheater (feed and hot-fluid sides)	R05	N2.4, N2.5	Verify PSV-101 is sized for blocked-in feed with hot fluid flowing (thermal expansion and vaporisation) and for E-100 tube rupture (two-thirds rule).	Process		OPEN	Risk 8 Medium
N2 E-100 feed preheater (feed and hot-fluid sides)	R06	N2.1, N2.3, N2.6	Add a hot-fluid return temperature indicator and an E-100 approach-temperature calculation so fouling is seen before TI-010 falls to its alarm.	Instrument		OPEN	Risk 9 Medium
N5/N6 Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun / Combustion air: F-100 intake screen → FV-200 → windbox and burner register	R07	N1.3, N2.1, N2.3, N3.3, N4.2, N5.3, N6.2, N7.5, N9.6	Implement cross-limiting (lead-lag) air/fuel control: fuel demand limited by measured air, air demand not below that needed for measured fuel.	Instrument		OPEN	Risk 12 High
N6 Combustion air: F-100 intake screen → FV-200 → windbox and burner register	R08	N5.3, N5.7, N5.8, N6.2, N6.3, N6.6, N6.7, N7.5, N8.5	Put the O ₂ trim (rev B) in service with a CO override: high CO blocks the trim from cutting air further.	Instrument		OPEN	Risk 12 High
N5 Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	R09	N5.2	Cascade TIC-030 to a fuel-flow controller on FI-001 so fuel-pressure disturbances are corrected before they reach the outlet temperature.	Instrument		OPEN	Risk 9 Medium
N4 H-100 process coil: convection → crossover → radiant → outlet to V-200	R10	N3.2, N4.3, N4.7	Add more radiant tube-skin thermocouples (hottest tubes) and evaluate a TI-024 high-high trip by LOPA.	Process / Instrument		OPEN	Risk 12 High
N4 H-100 process coil: convection → crossover → radiant → outlet to V-200	R11	N1.5, N4.4	Confirm coil design pressure ≥ pump shut-off head; otherwise add overpressure protection. Add high-pressure alarm on PI-005 (blocked outlet to V-200).	Process		OPEN	Risk 6 Medium

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N4 H-100 process coil: convection → crossover → radiant → outlet to V-200	R12	N3.1, N4.5, N5.9, N7.7	Provide remotely operated emergency isolation valves on the heater inlet and outlet, operable from the control room, for tube rupture.	Process		OPEN	Risk 10 High
N5 Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	R13	N5.6	Add fuel-oil temperature indication and low-temperature alarm; keep oil at atomising viscosity (tracing or heater).	Process		OPEN	Risk 9 Medium
N5 Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	R14	N5.1, N5.2, N5.5	Install a duplex strainer with ΔP indication upstream of FV-300.	Mechanical		OPEN	Risk 9 Medium
N5 Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	R15	N5.4, N9.1, N9.4, N9.9	Add proof-of-closure switches on XV-301/302 as purge permissives and a valve-leak test before every light-off.	Instrument		OPEN	Risk 10 High
N5 Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	R16	N1.9, N5.7, N7.1	Water-draw and sampling procedure for the fuel-oil tank; alarm on water in fuel if available.	Operations		OPEN	Risk 9 Medium
N8 Convection section, stack and damper HV-400	R17	N6.1, N6.5, N7.4, N8.1	Fit a mechanical minimum-open stop on damper HV-400 (e.g. 30 %) so it cannot close fully.	Mechanical		OPEN	Risk 8 Medium
N7/N8 Firebox (radiant section), burner and burner management system / Convection section, stack and damper HV-400	R18	N6.1, N7.4, N8.1	Evaluate a high firebox-pressure trip (PSHH on PI-003) by LOPA; today PI-003 only alarms.	Process / Instrument		OPEN	Risk 8 Medium
N7 Firebox (radiant section), burner and burner management system	R19	N6.7, N7.6	Add peep-door and explosion-door checks to the field operator round sheet; peep doors closed except when viewing.	Operations		OPEN	Risk 9 Medium
N7 Firebox (radiant section), burner and burner management system	R20	N7.1, N7.7, N7.8	Lock HV-221 snuffing steam closed while firing; drain condensate before use; write it into the fire-emergency procedure.	Operations		OPEN	Risk 10 High
N7 Firebox (radiant section), burner and burner management system	R21	N4.5, N7.2	Evaluate a high firebox-temperature trip (TSHH on TIAH-023) by LOPA.	Process		OPEN	Risk 10 High
N6/N7 Combustion air: F-100 intake screen → FV-200 → windbox and burner register / Firebox (radiant section), burner and burner management system	R22	N1.3, N2.3, N4.2, N5.3, N6.2, N6.6, N7.5, N8.3, N8.5, N9.6	Written response procedure for high CO (AI-026) and low O ₂ (AI-028): raise air first, then cut fuel; train on the simulator.	Operations		OPEN	Risk 12 High
N9 Operating procedures: start-up (purge, light-off) and shutdown	R23	N7.9, N9.1, N9.2, N9.8	Trip-bypass management: any BMS bypass needs a permit, time limit and MOC; bypass status shown on the DCS.	Operations		OPEN	Risk 10 High
N4 H-100 process coil: convection → crossover → radiant → outlet to V-200	R24	N4.1, N4.8	Add a deviation alarm TT-040 vs TIAH-025 (outlet transmitter disagreement).	Instrument		OPEN	Risk 9 Medium

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N6 Combustion air: F-100 intake screen → FV-200 → windbox and burner register	R25	N6.2, N6.4	Check FV-200 and F-100 capacity at the hottest ambient and highest firing; add intake-screen ΔP indication.	Process		OPEN	Risk 12 High
N8 Convection section, stack and damper HV-400	R26	N8.4	Low stack-temperature alarm on TI-021 when firing sulfur-bearing oil (acid dew point).	Instrument		OPEN	Risk 4 Low
N9 Operating procedures: start-up (purge, light-off) and shutdown	R27	N1.10, N9.3, N9.4, N9.5, N9.7	Start-up and shutdown procedures reviewed against this HAZOP; operators qualified on the simulator before first light-off.	Operations		OPEN	Risk 8 Medium

HAZOP REPORT

Hazard and Operability Study of the H-100 natural draft fired heater
Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator

	Role	Name	Signature	Date
Prepared by	HAZOP scribe			
Reviewed by	HAZOP leader (chair)			
Approved by	Plant / project manager			

This report records a teaching HAZOP of the heater drawn in the simulator. Names, signatures and dates are filled in by the class team that runs the study; the risk matrix is an example for teaching, not a corporate matrix.

CONTENTS

HAZOP CLOSE-OUT SUMMARY	2
Executive summary	7
1 Introduction	8
1.1 Purpose	8
1.2 The unit	8
1.3 Objectives	8
2 Scope	8
2.1 Nodes	8
2.2 Node map	9
3 Methodology	10
3.1 References	10
3.2 The four phases	10
3.3 Guide words and parameters	10
3.4 Risk ranking	11
4 HAZOP sessions	12
4.1 Schedule	12
4.2 Team	12
4.3 Documents reviewed	12
4.4 Nodes reviewed and guide-word coverage	12
5 Findings and recommendations	13
5.1 Risk profile	13
5.2 Recommendations	13
5.3 Dynamic check of selected deviations on the simulator	14
5.4 Follow-up and close-out	15
Appendix 1 • HAZOP worksheets	16
Node N1 • Feed supply: TK-101 → P-100A/B → E-100 inlet	16
Node N2 • E-100 feed preheater (feed and hot-fluid sides)	17
Node N3 • Feed flow control FIC-003 / FV-010 to the coil inlet	18
Node N4 • H-100 process coil: convection → crossover → radiant → outlet to V-200	19
Node N5 • Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	20
Node N6 • Combustion air: F-100 intake screen → FV-200 → windbox and burner register	21
Node N7 • Firebox (radiant section), burner and burner management system	22
Node N8 • Convection section, stack and damper HV-400	23
Node N9 • Operating procedures: start-up (purge, light-off) and shutdown	24
Appendix 2 • HAZOP documents (drawings with node boundaries)	25
A2-1 • HAZOP node map	25
A2-2 • P&ID-001 rev C (screen drawing used for the study)	26
A2-3 • NDFH-V1-PID-001 rev D (alternative EPC drawing)	27
Appendix 3 • HAZOP action close-out sheets	28

Executive summary

A HAZOP of the H-100 natural draft fired heater, from the feed tank TK-101 to the inlet of the packed-bed reactor V-200, including the fuel oil and combustion air systems, the firebox, convection section and stack, the burner management system (BMS) and the start-up and shutdown procedures, was carried out with the IEC 61882 guide-word method. The heater was divided into **9 nodes**; **69 deviations** were recorded and **27 recommendations** raised.

Nodes	Deviations	Recommendations	Very high	High	Medium	Low
9	69	27	0	9	50	10

Risk is ranked with the existing safeguards credited (Section 3.4). No deviation was ranked very high; the high-ranked deviations are those where an instrumented trip is the main protection, and they are carried to the LOPA (tab LOPA, SRS of SIF).

Key findings

N1 loss of feed: FSLL-008 trips within seconds, but the tube skin passes its 430 °C alarm first. The trip works; the margin is short, so the spare-pump auto-start (R01) matters.

N6 air meter reading high: no trip at all. O₂ goes to zero, CO rises to thousands of ppm and the outlet temperature falls. Only alarms and the operator protect, which is why R07, R08 and R22 were raised.

N5 fuel valve failing high with air unchanged: the outlet temperature *falls*, because the box goes fuel-rich. "More fuel" is a combustion hazard before it is an over-temperature hazard.

N8 damper stuck nearly closed: the firebox goes positive while the windbox draft stays above the PSLL-030 trip point. Nothing trips (R17, R18).

N4 outlet transmitter reading low: the real outlet reaches about 367 °C; TIAH-025 alarms but stays below the TSHH-025 trip (R24).

Actions by action party

Action party	Number	Action No.
Instrument	11	R01, R02, R06, R07, R08, R09, R10, R15, R18, R24, R26
Process	9	R03, R05, R10, R11, R12, R13, R18, R21, R25
Operations	7	R01, R16, R19, R20, R22, R23, R27
Mechanical	3	R04, R14, R17

An action shared by two parties (e.g. Instrument / Operations) is counted under both.

1 Introduction

1.1 Purpose

The HAZOP identifies how the H-100 heater can deviate from its design intent, what the deviations lead to, whether the existing safeguards are adequate, and what should be changed. It also finds operability problems: deviations that do not harm people but stop the heater or upset the reactor downstream. The study feeds the safety layers, LOPA and SRS of the SIFs, and the cause and effect diagram NDFH-V1-IN-CE-001.

1.2 The unit

H-100 is a natural draft, single-burner, oil-fired heater. Liquid hydrocarbon feed from TK-101 is pumped by P-100A (P-100B spare), preheated in E-100 by a heating fluid from Unit A, flows under FIC-003 through the convection and radiant coils, and leaves at about 350 °C to the packed-bed reactor V-200. The outlet temperature TIC-030 moves the fuel valve FV-300; combustion air is drawn in by the stack draft through F-100 and FV-200 under FIC-005, and the stack damper HV-400 sets the draft. SIS-100 / BMS trips the burner on low feed flow, high outlet temperature, low fuel pressure, loss of draft and flame failure.

1.3 Objectives

- Identify hazards to people, the environment and the plant from deviations of each node.
- Identify operability problems and their effect on V-200 and Unit A.
- Judge the existing safeguards and rank the risk with the example matrix.
- Raise recommendations and pass the high-ranked scenarios to LOPA.
- Check selected deviations dynamically on the simulator model (Section 5.3).

2 Scope

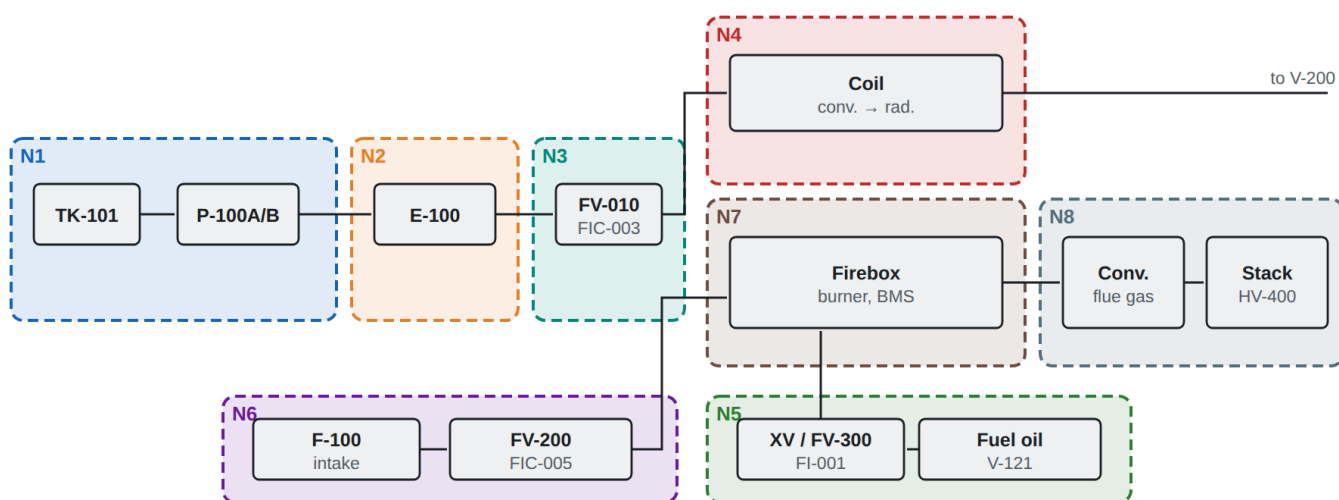
Definition: scope (H-100 from TK-101 to the V-200 inlet, fuel and air systems, stack, BMS; excluding V-200 and Unit A), objectives, team, risk matrix.

2.1 Nodes

Node	Description	Design intent	Design conditions	Equipment / instruments
N1	Feed supply: TK-101 → P-100A/B → E-100 inlet	Transfer liquid hydrocarbon feed from TK-101 at 40 °C with P-100A (P-100B spare) at about 1030 kPa discharge, 6.8 L/s, to E-100.	40 °C; suction ≈ 50 kPa; discharge ≈ 1030 kPa; 6.8 L/s	TK-101, P-100A/B, suction and discharge piping <i>PI-001, PI-002</i>
N2	E-100 feed preheater (feed and hot-fluid sides)	Heat feed from 40 to 180 °C (1.9 MW) with hot fluid from Unit A entering at 230 °C through HV-063 (60 %).	Feed 40 → 180 °C; hot fluid in 230 °C; HV-063 60 %	E-100 shell-and-tube, HV-063, PSV-101 <i>TI-063, TI-010, PSV-101</i>
N3	Feed flow control FIC-003 / FV-010 to the coil inlet	Control the feed to the heater at 6.8 L/s (FV-010, fail open) and protect the coil with FSLL-008.	6.8 L/s, FV-010 ≈ 57 %, ≈ 180 °C	FV-010, flow element, piping to the convection inlet <i>FIC-003, FIAL-008, FSLL-008</i>
N4	H-100 process coil: convection → crossover → radiant → outlet to V-200	Heat 5.44 kg/s of liquid feed from 180 to 350 °C (2.31 MW) in a single pass, tube skin below 430 °C, and deliver it to V-200 at about 545 kPa.	180 → 233 (crossover) → 350 °C; skin ≈ 398 °C; ≈ 545 kPa outlet	Convection tubes, crossover, radiant tubes, outlet line <i>TI-020, TI-022, TI-024, TT-040/TIC-030, TIAH-025/TSHH-025, PI-005</i>
N5	Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	Deliver 243 kg/h fuel oil at the rate TIC-030 demands (FV-300 fail closed), atomised at about 600 kPa at the gun; shut off tightly on any trip.	Header ≈ 1940 kPa; gun ≈ 600 kPa; FV-300 ≈ 45 %	Isolation V-121, V-122; double block and bleed XV-301/XV-303/XV-302; FV-300; gun <i>FI-001, PI-019, PIAL-020 / PSLL-020, TIC-030 output</i>
N6	Combustion air: F-100 intake screen → FV-200 → windbox and burner register	Supply 3892 kg/h combustion air ($\lambda \approx 1.11$, firebox $O_2 \approx 2$ %) by natural draft, controlled by FIC-005 through FV-200 (fail open).	3892 kg/h; FV-200 ≈ 48 %; windbox draft ≈ 5.25 mmH ₂ O	F-100 screen, FV-200, windbox, air register <i>FIC-005, FY-005 (upgrade), AIC-027 (rev B), PIAL-030 / PSLL-030</i>

Node	Description	Design intent	Design conditions	Equipment / instruments
N7	Firebox (radiant section), burner and burner management system	Burn the fuel completely in a stable flame at slightly negative pressure; transfer ~1.6 MW by radiation with bridgwall $\approx 990^\circ\text{C}$; BMS purges, proves flame and trips on unsafe conditions.	$\approx 990^\circ\text{C}$; $-2.5\text{ mmH}_2\text{O}$ at the arch; $\text{O}_2 \approx 2\%$	Burner, flame scanner BS-005, refractory, explosion doors, peep doors, snuffing steam HV-221 <i>TIAH-023, PI-003, AI-028, BS-005, BMS</i>
N8	Convection section, stack and damper HV-400	Cool flue gas from ≈ 990 to 400°C over the convection tubes and discharge it; produce $\approx -2.5\text{ mmH}_2\text{O}$ at the arch with damper HV-400 at 67 % (fail open).	Stack 400°C ; $\text{O}_2 \approx 3.2\%$; $\text{CO} \approx 20\text{ ppm}$	Convection bank, stack, damper HV-400 <i>TI-021, PI-031, PI-032, AI-026, AI-029, HIC-400</i>
N9	Operating procedures: start-up (purge, light-off) and shutdown	Start: feed flow established, draft set, 5-min purge (air $\geq 1700\text{ kg/h}$, damper $\geq 60\%$), light-off at 10–20 % fuel with a 10-s trial, then ramp to 350°C . Shutdown: cut fuel, burner out, keep feed until cool.	Procedure steps in the Exercises tab	BMS sequence, operator actions <i>BMS screen, FIC-003, FIC-005, TIC-030 (MAN), PI-004</i>

2.2 Node map



N9 (not drawn): start-up, shutdown and purge procedures

Figure 1. H-100 divided into HAZOP nodes (N9, the procedures node, covers the whole heater).

3 Methodology

HAZOP is a structured, team-based examination of a defined system. Each part of the design is compared with its **design intent**; **guide words** combined with process **parameters** generate **deviations**; for each credible deviation the team records causes, consequences, existing safeguards and, where the risk is not acceptable, recommendations. It finds both hazards (harm to people, environment, assets) and operability problems (the plant cannot make product or be run as intended).

3.1 References

Reference	What it contributes
IEC 61882:2016 <i>Hazard and operability studies (HAZOP studies) – Application guide</i>	Definitions, the four phases (definition, preparation, examination, documentation and follow-up), guide words, team roles, recording
CCPS <i>Guidelines for Hazard Evaluation Procedures</i> (3rd ed.)	Practical worksheet format, node selection, "consequence without safeguards", risk ranking, links to LOPA
IEC 61511 (functional safety, SIS)	How HAZOP scenarios feed LOPA and SIL targets for trips such as TSHH-025 or a proposed firebox-pressure trip
ISO 31000 / owner's risk matrix	Risk criteria used to decide whether safeguards are adequate
API 560, API 556, NFPA 85/86	Fired-heater design, instrumentation, burner management and purge practice used as the basis for safeguards
Thailand: Department of Industrial Works regulation on hazard identification, risk assessment and risk-management plans (B.E. 2543)	Legal requirement to identify hazards (HAZOP is an accepted method) and to prepare risk-management plans for factories
OSHA 29 CFR 1910.119 (PSM), where applicable	Process hazard analysis for covered processes, revalidated at least every five years

3.2 The four phases

- **Definition:** scope (H-100 from TK-101 to the V-200 inlet, fuel and air systems, stack, BMS; excluding V-200 and Unit A), objectives, team, risk matrix.
- **Preparation:** collect the P&ID, PFD, heat and material balance, cause-and-effect (trip) chart, operating procedures, alarm list, datasheets; divide the plant into nodes; write each node's design intent; choose parameters and guide words.
- **Examination:** for each node → state the design intent → choose a parameter → apply each guide word → is the deviation meaningful? → causes → consequences assuming the safeguards fail → existing safeguards → rank the risk → recommend → next guide word. Record all meaningful deviations, including those with adequate safeguards ("full recording"), so the reasoning can be audited.
- **Documentation and follow-up:** worksheets, recommendation register with owners and dates, action close-out, and revalidation after changes (MOC) or periodically.

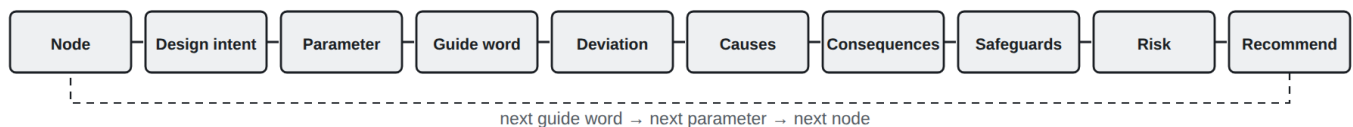


Figure 2. HAZOP examination loop.

3.3 Guide words and parameters

Guide word	Meaning (IEC 61882)
NO / NOT	Complete negation of the intent
MORE	Quantitative increase
LESS	Quantitative decrease
AS WELL AS	Qualitative increase: something extra present
PART OF	Qualitative decrease: only part of the intent achieved
REVERSE	Logical opposite of the intent
OTHER THAN	Complete substitution: something else happens
EARLY	Relative to clock time
LATE	Relative to clock time
BEFORE	Relative to order or sequence
AFTER	Relative to order or sequence

Parameters used: flow, pressure, temperature (process and tube metal), level, composition, heat transfer, flame, integrity (containment), measurement, control, utility, safety system, maintenance, and for procedures: purge, light-off, sequence, relight, shutdown. A deviation is only recorded if it is physically meaningful ("REVERSE temperature" is not).

3.4 Risk ranking

S \ L	L1	L2	L3	L4	L5
S5	5 (6)	10 (4)	15	20	25
S4	4 (4)	8 (13)	12 (5)	16	20
S3	3	6 (8)	9 (17)	12	15
S2	2	4 (5)	6 (5)	8 (1)	10
S1	1 (1)	2	3	4	5

Numbers in brackets: how many deviations of this study fall in the cell. Risk = S × L.

S	Severity (worst credible consequence)
1	First aid; minor damage; no release
2	Medical treatment; short outage; small contained release
3	Lost-time injury; heater shutdown for days; local release
4	Permanent disability or single fatality; major heater damage; fire
5	Multiple fatalities; firebox explosion; fire spreading beyond the unit

L	Likelihood with existing safeguards
1	Remote: not expected in the life of the plant ($< 10^{-4}$ /y)
2	Unlikely: known in industry (10^{-4} - 10^{-3} /y)
3	Possible: could happen in the plant life (10^{-3} - 10^{-2} /y)
4	Likely: several times in the plant life (10^{-2} - 10^{-1} /y)
5	Frequent: yearly or more

1-4 Low acceptable · 5-9 Medium reduce if practicable (ALARP), review safeguards · 10-16 High recommendation required, consider LOPA · 20-25 Very high not acceptable, act before operation

4 HAZOP sessions

4.1 Schedule

The study is run in class as a teaching HAZOP. A typical plan is three sessions; the team records the dates and attendance on this page.

Session	Nodes	Date	Remark
1	N1 – N4 (feed, preheat, flow control, coil)		
2	N5 – N7 (fuel, air, firebox and BMS)		
3	N8 – N9 (stack, procedures); review of recommendations		

4.2 Team

Role	Responsibility	Name	Attended sessions
HAZOP leader (chair)	Independent of the design; runs the method, keeps the team on the node, decides when a deviation is covered		
Scribe (recorder)	Writes the worksheet live, reads back recommendations		
Process engineer (designer)	Explains the design intent, balances and limits		
Operations representative	How the heater is really run: start-up, upsets, rounds, procedures		
Instrument / control engineer	Loops, fail positions, BMS logic, trip independence		
Safety / risk engineer	Risk matrix, LOPA follow-up, regulations		
Specialists as needed	Combustion / burner vendor, mechanical (tubes, refractory), maintenance		

4.3 Documents reviewed

Document	Title / content
P&ID-001 rev C	H-100 P&ID, screen drawing of the simulator (Appendix 2)
NDFH-V1-PID-001 rev D	H-100 P&ID, alternative EPC-standard drawing (Appendix 2)
PFD-100	Process flow diagram with the stream table
NDFH-V1-IN-CE-001 rev R0	Cause and effect diagram of SIS-100 / BMS
NDFH-V1-IN-SRS-001 rev R0	Safety requirements specification of the SIFs
Simulator tabs	Process, Design, Control loop design, alarm and trip list (Workbook tab), start-up and shutdown procedures (SOP tab)

4.4 Nodes reviewed and guide-word coverage

Node	NO / NOT	MORE	LESS	AS WELL AS	PART OF	REVERSE	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows
N1	1	3	3	1	·	1	1	·	·	·	·	10
N2	1	2	2	·	·	·	1	·	·	·	·	6
N3	1	1	1	·	·	1	1	·	·	·	·	5
N4	·	3	2	1	1	·	1	·	·	·	·	8
N5	1	2	2	1	·	·	3	·	·	·	·	9
N6	1	2	2	·	·	·	2	·	·	·	·	7
N7	1	3	1	2	·	·	2	·	·	·	·	9
N8	·	4	2	·	·	·	·	·	·	·	·	6
N9	1	1	1	·	·	·	1	1	1	2	1	9

A dot means that guide word gave no meaningful deviation for any parameter in the node (it was still considered). EARLY / LATE / BEFORE / AFTER apply mainly to the procedure node N9.

5 Findings and recommendations

5.1 Risk profile

Node	Deviations	High	Medium	Low	Recommendations
N1 Feed supply: TK-101 → P-100A/B → E-100 inlet	10	0	9	1	R01, R02, R03, R04
N2 E-100 feed preheater (feed and hot-fluid sides)	6	0	5	1	R05, R06
N3 Feed flow control FIC-003 / FV-010 to the coil inlet	5	0	4	1	
N4 H-100 process coil: convection → crossover → radiant → outlet to V-200	8	2	5	1	R10, R11, R12, R24
N5 Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	9	1	8	0	R07, R09, R13, R14, R15, R16
N6 Combustion air: F-100 intake screen → FV-200 → windbox and burner register	7	1	6	0	R07, R08, R22, R25
N7 Firebox (radiant section), burner and burner management system	9	3	5	1	R18, R19, R20, R21, R22
N8 Convection section, stack and damper HV-400	6	1	4	1	R17, R18, R26
N9 Operating procedures: start-up (purge, light-off) and shutdown	9	1	4	4	R23, R27
Total	69	9	50	10	27

High-ranked deviations

Ref	Deviation	Consequence	S×L	Rec.
N4.3	Temperature (tube metal) MORE: Tube skin above limit	Creep damage, bulging, tube rupture → fire	4×3 = 12	R10
N4.5	Pressure LESS: Loss of containment	Hydrocarbon burns in the firebox: firebox temperature and pressure rise, smoke, possible explosion	5×2 = 10	R12, R21
N5.4	Flow AS WELL AS: Fuel leaks into the firebox while shut down	Fuel accumulates; explosion at the next light-off	5×2 = 10	R15
N6.2	Flow LESS: Less air than needed (λ → 1 or below)	CO and smoke; outlet temperature falls although fuel rises; afterburning in convection; explosion risk	4×3 = 12	R07, R08, R22, R25
N7.5	Composition MORE: Too much combustibles (CO)	Afterburning, soot, explosion risk	4×3 = 12	R07, R08, R22
N7.7	Composition AS WELL AS: Hydrocarbon from a leaking tube	Uncontrolled fire in the box, high temperature, smoke	5×2 = 10	R12, R20
N7.9	Safety system OTHER THAN: BMS trip bypassed	Unsafe condition goes unprotected	5×2 = 10	R23
N8.5	Composition MORE: High CO at the stack	Toxic emission, afterburning, explosion risk in the convection section	4×3 = 12	R08, R22
N9.6	Sequence BEFORE: Fuel raised before air (load increase)	Fuel-rich firing, CO	4×3 = 12	R07, R22

5.2 Recommendations

No.	Recommendation	Node	Raised from	Action by
R01	Auto-start spare P-100B on low discharge pressure (PI-002) or P-100A trip, with the start completing inside the FSLL-008 delay; test the spare monthly.	N1	N1.1	Instrument / Operations
R02	Add level transmitter LT-101 on TK-101 with low-level alarm LAL and low-low pump trip LSLL to protect P-100A/B from running dry.	N1	N1.1, N1.2, N1.6, N1.7, N1.8	Instrument
R03	Provide minimum-flow recirculation (restriction orifice back to TK-101) on P-100A/B so a closed FV-010 cannot deadhead the running pump.	N1	N1.5	Process
R04	Confirm a check valve on each pump discharge; include in the inspection plan (passing check valve spins the idle pump backwards).	N1	N1.4, N3.4	Mechanical
R05	Verify PSV-101 is sized for blocked-in feed with hot fluid flowing (thermal expansion and vaporisation) and for E-100 tube rupture (two-thirds rule).	N2	N2.4, N2.5	Process
R06	Add a hot-fluid return temperature indicator and an E-100 approach-temperature calculation so fouling is seen before TI-010 falls to its alarm.	N2	N2.1, N2.3, N2.6	Instrument

No.	Recommendation	Node	Raised from	Action by
R07	Implement cross-limiting (lead-lag) air/fuel control: fuel demand limited by measured air, air demand not below that needed for measured fuel.	N5/N6	N1.3, N2.1, N2.3, N3.3, N4.2, N5.3, N6.2, N7.5, N9.6	Instrument
R08	Put the O ₂ trim (rev B) in service with a CO override: high CO blocks the trim from cutting air further.	N6	N5.3, N5.7, N5.8, N6.2, N6.3, N6.6, N6.7, N7.5, N8.5	Instrument
R09	Cascade TIC-030 to a fuel-flow controller on FI-001 so fuel-pressure disturbances are corrected before they reach the outlet temperature.	N5	N5.2	Instrument
R10	Add more radiant tube-skin thermocouples (hottest tubes) and evaluate a TI-024 high-high trip by LOPA.	N4	N3.2, N4.3, N4.7	Process / Instrument
R11	Confirm coil design pressure $\geq$ pump shut-off head; otherwise add overpressure protection. Add high-pressure alarm on PI-005 (blocked outlet to V-200).	N4	N1.5, N4.4	Process
R12	Provide remotely operated emergency isolation valves on the heater inlet and outlet, operable from the control room, for tube rupture.	N4	N3.1, N4.5, N5.9, N7.7	Process
R13	Add fuel-oil temperature indication and low-temperature alarm; keep oil at atomising viscosity (tracing or heater).	N5	N5.6	Process
R14	Install a duplex strainer with ΔP indication upstream of FV-300.	N5	N5.1, N5.2, N5.5	Mechanical
R15	Add proof-of-closure switches on XV-301/302 as purge permissives and a valve-leak test before every light-off.	N5	N5.4, N9.1, N9.4, N9.9	Instrument
R16	Water-draw and sampling procedure for the fuel-oil tank; alarm on water in fuel if available.	N5	N1.9, N5.7, N7.1	Operations
R17	Fit a mechanical minimum-open stop on damper HV-400 (e.g. 30 %) so it cannot close fully.	N8	N6.1, N6.5, N7.4, N8.1	Mechanical
R18	Evaluate a high firebox-pressure trip (PSHH on PI-003) by LOPA; today PI-003 only alarms.	N7/N8	N6.1, N7.4, N8.1	Process / Instrument
R19	Add peep-door and explosion-door checks to the field operator round sheet; peep doors closed except when viewing.	N7	N6.7, N7.6	Operations
R20	Lock HV-221 snuffing steam closed while firing; drain condensate before use; write it into the fire-emergency procedure.	N7	N7.1, N7.7, N7.8	Operations
R21	Evaluate a high firebox-temperature trip (TSHH on TIAH-023) by LOPA.	N7	N4.5, N7.2	Process
R22	Written response procedure for high CO (AI-026) and low O ₂ (AI-028): raise air first, then cut fuel; train on the simulator.	N6/N7	N1.3, N2.3, N4.2, N5.3, N6.2, N6.6, N7.5, N8.3, N8.5, N9.6	Operations
R23	Trip-bypass management: any BMS bypass needs a permit, time limit and MOC; bypass status shown on the DCS.	N9	N7.9, N9.1, N9.2, N9.8	Operations
R24	Add a deviation alarm TT-040 vs TIAH-025 (outlet transmitter disagreement).	N4	N4.1, N4.8	Instrument
R25	Check FV-200 and F-100 capacity at the hottest ambient and highest firing; add intake-screen ΔP indication.	N6	N6.2, N6.4	Process
R26	Low stack-temperature alarm on TI-021 when firing sulfur-bearing oil (acid dew point).	N8	N8.4	Instrument
R27	Start-up and shutdown procedures reviewed against this HAZOP; operators qualified on the simulator before first light-off.	N9	N1.10, N9.3, N9.4, N9.5, N9.7	Operations

5.3 Dynamic check of selected deviations on the simulator

A conventional HAZOP judges consequences qualitatively. Here selected deviations are run on a copy of the H-100 model from design conditions with the existing safeguards active (alarms, BMS trips, controllers in their normal modes, no operator action) for up to 30 min. The result shows whether the safeguards act before harm and how much time the operator has.

Ref	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trip (first-out)	Range reached before trip / in 30 min
N1	Flow NO	Running pump P-100A trips (spare not started)	FIAL-008 LO at 00:01 TI-024 HI at 00:07	FSLL-008 low process flow at 00:07	FIC-003 0.59 to 6.80 L/s TIAH-025 351.3 to 355.5 °C TI-024 398 to 432 °C

Ref	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trip (first-out)	Range reached before trip / in 30 min
N4	Temperature MORE	TT-040 reads 15 °C low (TIC-030 over-fires)	TIC-030 LO at 00:26 AI-028 LO at 01:01 TIAH-025 HI at 02:11	no trip	TIAH-025 351.3 to 367.0 °C TI-024 398 to 419 °C TT-040 337.4 to 350.7 °C
N5	Flow MORE	TIC-030 output fails 25 % high, air unchanged	AI-026 HI at 00:03 AI-028 LO at 00:07 AI-029 LO at 00:18	no trip	TIAH-025 338.4 to 351.3 °C AI-028 0.00 to 2.10 % O ₂ AI-026 20 to 2619 ppm
N6	Flow LESS	Air meter FT-005 reads 50 % high	AI-028 LO at 00:13 AI-026 HI at 00:14 AI-029 LO at 00:51	no trip	AI-028 0.00 to 2.10 % O ₂ AI-026 20 to 7781 ppm TT-040 302.4 to 350.2 °C
N2	Temperature LESS	E-100 fouled (full severity, 5 min ramp)	TI-010 LO at 01:52 TIC-030 LO at 04:16 AI-028 LO at 04:39	no trip	TI-010 131.0 to 180.0 °C TI-024 293 to 398 °C AI-028 0.00 to 2.10 % O ₂
N5	Flow LESS	Fuel supply pressure falls	TIC-030 LO at 00:59 AI-028 HI at 01:23 AI-029 HI at 01:32	no trip	PIAL-020 235 to 600 kPa TT-040 303.7 to 350.0 °C
N8	Flow LESS	Damper HV-400 sticks nearly closed	PI-003 HI at 00:43 AI-028 LO at 00:45 AI-026 HI at 00:50	no trip	PI-003 -2.50 to 2.27 mmH ₂ O PIAL-030 0.77 to 5.25 mmH ₂ O AI-028 0.00 to 2.10 % O ₂
N7	Flame NO	Flame failure	none in 30 min	BSLL-005 flame failure at 00:02	TT-040 350.0 to 350.0 °C
N4	Pressure LESS	Radiant tube leak	AI-028 LO at 00:51 TIC-030 HI at 01:47	no trip	TIAH-023 982 to 1013 °C PI-003 -2.50 to -2.45 mmH ₂ O AI-028 0.73 to 2.12 % O ₂

5.4 Follow-up and close-out

- Each recommendation gets an owner and due date; the owner answers with *accepted and done*, *accepted with alternative* (with justification) or *rejected* (with justification agreed by the HAZOP leader and plant manager).
- High and very-high scenarios whose protection relies on instrumented trips go to **LOPA** to set the required SIL (IEC 61511).
- Any change (e.g. putting rev B in service, adding cross-limiting) goes through **management of change**, which checks whether the HAZOP must be updated.
- Revalidate the HAZOP periodically (typically every five years) and after incidents.
- Use the simulator to train operators on the scenarios the HAZOP found; verify that procedures and alarm responses work.

Appendix 1 · HAZOP worksheets

Node N1 · Feed supply: TK-101 → P-100A/B → E-100 inlet

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Node	N1 · Feed supply: TK-101 → P-100A/B → E-100 inlet
Design intent	Transfer liquid hydrocarbon feed from TK-101 at 40 °C with P-100A (P-100B spare) at about 1030 kPa discharge, 6.8 L/s, to E-100.	Design conditions	40 °C; suction ≈ 50 kPa; discharge ≈ 1030 kPa; 6.8 L/s
Equipment	TK-101, P-100A/B, suction and discharge piping	Instruments	PI-001, PI-002
Drawing	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Document	NDFH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.1	Flow	NO	No feed flow to E-100	- Running pump P-100A trips (motor, power) - suction valve closed - TK-101 empty - P-100B fails to start	Coil loses its coolant while firing → tube overheating, coking, rupture and fire in the firebox	- FSL-008 trips the burner (< 3.4 L/s, 5 s) - FIAL-008 LO alarm - P-100B spare (manual start)	4	2	8 M	R01, R02	Instrument · Instrument / Operations
N1.2	Flow	LESS	Less feed flow	- Pump wear or cavitation - partly closed suction valve - suction strainer plugged - low TK-101 level	Higher coil outlet temperature and skin temperature at the same firing; production shortfall	- TIC-030 cuts fuel - FIAL-008 LO - FSL-008 at 3.4 L/s	3	3	9 M	R02	Instrument
N1.3	Flow	MORE	More feed flow	- Operator raises FIC-003 SP - FV-010 fails open (FO)	More duty required; fuel rises; air may not follow (rev A) → fuel-rich firebox, CO, smoke	- TIC-030 - AI-028 O₂ LO alarm - AI-026 CO HI alarm - operator raises air first	3	3	9 M	R07, R22	Instrument · Operations
N1.4	Flow	REVERSE	Reverse flow through the idle pump	Passing check valve on P-100B discharge	Recirculation through the idle pump, less flow forward, idle pump spins backwards	- Check valves - FIAL-008 LO	2	2	4 L	R04	Mechanical
N1.5	Pressure	MORE	High discharge pressure	- FV-010 or downstream valve closed (deadhead) - V-200 blocked	Pump overheats at no flow; possible seal failure and leak (fire)	- Pump shut-off head below piping design pressure - PI-002	3	2	6 M	R03, R11	Process
N1.6	Pressure	LESS	Low suction pressure	- Low TK-101 level - plugged suction strainer - hot feed near bubble point	Cavitation, pump damage, flow loss → see NO flow	- PI-001 (local) - operator rounds	3	3	9 M	R02	Instrument
N1.7	Level	LESS	Low level in TK-101	- Supply from upstream stopped - tank drained	Pumps lose suction → no flow	Operator monitoring only (no level instrument on the P&ID)	3	3	9 M	R02	Instrument
N1.8	Level	MORE	High level in TK-101	Upstream supply more than consumption	Tank overflow, spill, fire risk (outside this node)	Tank overflow line (upstream system)	3	2	6 M	R02	Instrument
N1.9	Composition	AS WELL AS	Water in the feed	Tank bottoms water carried over	Water flashes in the coil: pressure surges, hammer, flow upset, TIC-030 disturbance	None specific	3	2	6 M	R16	Operations
N1.10	Maintenance	OTHER THAN	Pump change-over done incorrectly	Spare started with discharge valve closed, or running pump stopped first	Flow interruption → FSL-008 trip of the heater	- Procedure - FSL-008	2	3	6 M	R27	Operations

Node N2 · E-100 feed preheater (feed and hot-fluid sides)

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Node	N2 · E-100 feed preheater (feed and hot-fluid sides)
Design intent	Heat feed from 40 to 180 °C (1.9 MW) with hot fluid from Unit A entering at 230 °C through HV-063 (60 %).	Design conditions	Feed 40 → 180 °C; hot fluid in 230 °C; HV-063 60 %
Equipment	E-100 shell-and-tube, HV-063, PSV-101	Instruments	TI-063, TI-010, PSV-101
Drawing	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Document	NDFH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.1	Temperature	LESS	Feed to H-100 colder than 180 °C	- HV-063 closed or Unit A cuts hot fluid - E-100 fouling - hot fluid colder	Heater must fire harder: more fuel, higher flux and tube skin; outlet may droop if air does not follow	- TI-010 LO alarm 170 °C - TIC-030 adds fuel - TI-024 HI 430 °C	3	3	9 M	R06, R07	Instrument
N2.2	Temperature	MORE	Feed hotter than design	- HV-063 opened fully - hotter hot fluid	Less firing needed; low-fire stability of the burner; possible vaporisation if pressure is low	- TIC-030 cuts fuel - burner turndown	2	2	4 L	—	—
N2.3	Flow (hot fluid)	NO	No hot fluid	- Unit A trip - HV-063 closed	Feed enters at about 40 °C: firing needed nearly doubles, beyond air/draft capacity; outlet falls; possible fuel-rich firing	- TI-010 LO alarm - TIC-030 - AI-028, AI-026 alarms	3	3	9 M	R06, R07, R22	Instrument · Operations
N2.4	Pressure	MORE	Feed side blocked in with hot fluid flowing	Inlet and outlet feed valves closed during an outage	Thermal expansion / vaporisation → overpressure of the E-100 feed side	PSV-101	4	2	8 M	R05	Process
N2.5	Integrity	OTHER THAN	E-100 tube leak	Corrosion, vibration	Hot fluid and feed mix; contamination of either system; high-pressure side flows into the low-pressure side	- PSV-101 (if feed is the low-pressure side) - product quality checks	3	2	6 M	R05	Process
N2.6	Heat transfer	LESS	Fouling	Deposits on the feed or hot side	Gradual fall in TI-010 and rise in firing; efficiency loss	- TI-010 LO alarm - cleaning programme	2	4	8 M	R06	Instrument

Node N3 · Feed flow control FIC-003 / FV-010 to the coil inlet

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Node	N3 · Feed flow control FIC-003 / FV-010 to the coil inlet
Design intent	Control the feed to the heater at 6.8 L/s (FV-010, fail open) and protect the coil with FSLL-008.	Design conditions	6.8 L/s, FV-010 ≈ 57 %, ≈ 180 °C
Equipment	FV-010, flow element, piping to the convection inlet	Instruments	FIC-003, FIAL-008, FSLL-008
Drawing	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Document	NDFH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.1	Flow	NO	No flow to the coil	- FV-010 closed by controller fault or operator - line blocked	Tube overheating while firing → coking, rupture, firebox fire	- FSLL-008 trip (independent element, 5 s) - FIAL-008 LO - FV-010 fails open on air failure	4	2	8 M	R12	Process
N3.2	Flow	LESS	Low flow	- FIC-003 SP lowered too far - FT reads high	High skin and outlet temperatures; coking	- FIAL-008 LO 5.0 L/s - FSLL-008 3.4 L/s - TIC-030	3	3	9 M	R10	Process / Instrument
N3.3	Flow	MORE	High flow	- FT-003 reads low - FV-010 fails open	Duty above capacity; outlet temperature falls; fuel-rich firing if air lags	- TIC-030 - AI-028/AI-026 alarms	3	2	6 M	R07	Instrument
N3.4	Flow	REVERSE	Back-flow from the coil	Pump stops with coil at higher pressure (V-200 pressure)	Hot hydrocarbon returns toward E-100 and pumps	Pump discharge check valves	2	2	4 L	R04	Mechanical
N3.5	Instrument	OTHER THAN	Flow measurement wrong	Transmitter drift, plugged impulse line	Controller holds the wrong flow; the independent FSLL-008 still protects	Separate FSLL-008 element	3	2	6 M	—	—

Node N4 · H-100 process coil: convection → crossover → radiant → outlet to V-200

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Node	N4 · H-100 process coil: convection → crossover → radiant → outlet to V-200
Design intent	Heat 5.44 kg/s of liquid feed from 180 to 350 °C (2.31 MW) in a single pass, tube skin below 430 °C, and deliver it to V-200 at about 545 kPa.	Design conditions	180 → 233 (crossover) → 350 °C; skin ≈ 398 °C; ≈ 545 kPa outlet
Equipment	Convection tubes, crossover, radiant tubes, outlet line	Instruments	TI-020, TI-022, TI-024, TT-040/TIC-030, TIAH-025/TSHH-025, PI-005
Drawing	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Document	NDFH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.1	Temperature	MORE	Outlet above 350 °C	- TIC-030 fault or in MAN with high output - TT-040 reads low - feed flow falls - set point error	Cracking/coking in the coil, catalyst damage in V-200, tube creep	- TIC-030 HI 356 °C - TIAH-025 HI 360 °C - TSHH-025 trip 375 °C (independent element)	4	2	8 M	R24	Instrument
N4.2	Temperature	LESS	Outlet below 350 °C	- Air-limited (fuel-rich) firing - fuel supply low - nozzle plugged - feed colder or higher	Off-spec reaction in V-200; operator may add fuel into a fuel-rich box	- TIC-030 LO 344 °C - AI-028 O₂ LO - AI-026 CO HI	3	3	9 M	R07, R22	Instrument · Operations
N4.3	Temperature (tube metal)	MORE	Tube skin above limit	- Coke inside tubes - low flow - flame impingement - over-firing	Creep damage, bulging, tube rupture → fire	- TI-024 HI 430 °C - FSL-008 - TIC-030	4	3	12 H	R10	Process / Instrument
N4.4	Pressure	MORE	Coil overpressure	Outlet blocked (V-200 inlet valve closed) with pump running	Coil rupture if pump shut-off exceeds design pressure; fire	PI-005 indication only	5	1	5 M	R11	Process
N4.5	Pressure	LESS	Loss of containment	Tube rupture (corrosion, creep, overheating)	Hydrocarbon burns in the firebox: firebox temperature and pressure rise, smoke, possible explosion	- TIAH-023 HI - PI-003 HI - FSL-008 as flow drops - snuffing steam HV-221 - ESD	5	2	10 H	R12, R21	Process
N4.6	Flow	PART OF	Part of the coil starved (maldistribution)	- Not applicable for a single pass - would apply to multi-pass designs	—	—	1	1	1 L	—	—
N4.7	Composition	AS WELL AS	Coke builds up	High skin temperature, low velocity, off-spec feed	Rising skin temperature and pressure drop, lower heat pickup	- TI-024 trend - decoking programme	3	3	9 M	R10	Process / Instrument
N4.8	Measurement	OTHER THAN	Outlet transmitter wrong	TT-040 drift or failure	TIC-030 over- or under-fires; TIAH-025 disagrees	Independent TIAH-025 / TSHH-025	3	3	9 M	R24	Instrument

Node N5 · Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Node	N5 · Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun
Design intent	Deliver 243 kg/h fuel oil at the rate TIC-030 demands (FV-300 fail closed), atomised at about 600 kPa at the gun; shut off tightly on any trip.	Design conditions	Header ≈ 1940 kPa; gun ≈ 600 kPa; FV-300 ≈ 45 %
Equipment	Isolation V-121, V-122; double block and bleed XV-301/XV-303/XV-302; FV-300; gun	Instruments	FI-001, PI-019, PIAL-020 / PSLL-020, TIC-030 output
Drawing	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Document	NDFH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.1	Flow	NO	No fuel to the burner	- Supply pump trip - V-121 closed - XV closed - strainer blocked	Flame-out; outlet temperature falls; risk of relighting without purge	- PSLL-020 and BSLL-005 trips - BMS requires purge before relight	3	3	9 M	R14	Mechanical
N5.2	Flow	LESS	Less fuel than demanded	- Supply pressure falls - nozzle partly plugged - strainer fouling - FV-300 sticks	Outlet temperature falls; FV-300 opens to 100 %; low gun pressure → poor atomisation, unstable flame	- PIAL-020 LO 100 kPa - PSLL-020 trip 40 kPa - TIC-030 LO alarm	3	3	9 M	R09, R14	Instrument · Mechanical
N5.3	Flow	MORE	More fuel than needed	- TIC-030 fault - TT-040 reads low - operator in MAN - FV-300 sticks open	High outlet temperature; if air does not follow → fuel-rich firing, CO, afterburning, explosion risk	- TIC-030 HI - TSHH-025 - AI-028 LO - AI-026 CO HI - FV-300 FC	4	2	8 M	R07, R08, R22	Instrument · Operations
N5.4	Flow	AS WELL AS	Fuel leaks into the firebox while shut down	- XV-301/302 passing - bleed XV-303 blocked	Fuel accumulates; explosion at the next light-off	- Double block and bleed - BMS purge (5 min) before light-off	5	2	10 H	R15	Instrument
N5.5	Pressure	MORE	High fuel pressure at the gun	- Control valve open fully - plugged gun tip (pressure rises for same flow)	Flame lift-off / instability; high firing	PIAL-020 HI 1400 kPa	3	2	6 M	R14	Mechanical
N5.6	Temperature	LESS	Cold, viscous fuel oil	- Tracing or heater failure - cold weather	Poor atomisation, long smoky flame, CO, soot on convection tubes	- Operator observation - AI-026 CO HI	3	3	9 M	R13	Process
N5.7	Composition	OTHER THAN	Different fuel quality	- New fuel batch with different LHV - water in fuel	Changed air requirement; sputtering flame, flame-out (water)	- AI-028 O₂ - BSLL-005 flame failure trip	3	3	9 M	R08, R16	Instrument · Operations
N5.8	Measurement	OTHER THAN	FI-001 reads wrong	- Wrong orifice plate - drift	Ratio FY-005 (if used) sets wrong air; wrong efficiency calculations	AI-028 O ₂ as independent check	3	2	6 M	R08	Instrument
N5.9	Integrity	OTHER THAN	Fuel leak at the burner front	Gasket or hose failure	Pool fire at the heater floor	- Fire detection and fire fighting - V-121 isolation - ESD	4	2	8 M	R12	Process

Node N6 · Combustion air: F-100 intake screen → FV-200 → windbox and burner register

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Node	N6 · Combustion air: F-100 intake screen → FV-200 → windbox and burner register
Design intent	Supply 3892 kg/h combustion air ($\lambda \approx 1.11$, firebox $O_2 \approx 2\%$) by natural draft, controlled by FIC-005 through FV-200 (fail open).	Design conditions	3892 kg/h; FV-200 $\approx 48\%$; windbox draft ≈ 5.25 mmH ₂ O
Equipment	F-100 screen, FV-200, windbox, air register	Instruments	FIC-005, FY-005 (upgrade), AIC-027 (rev B), PIAL-030 / PSL-030
Drawing	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Document	NDFH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N6.1	Flow	NO	No combustion air	- FV-200 closed by fault - windbox draft lost (damper closed) - F-100 blocked	Flame goes fuel-rich then out; unburnt fuel in the box → explosion on re-ignition	- PSLL-030 trip (< 0.5 mmH₂O draft, 3 s) - BSLL-005 - FV-200 fails open	5	1	5 M	R17, R18	Mechanical · Process / Instrument
N6.2	Flow	LESS	Less air than needed ($\lambda \rightarrow 1$ or below)	- Fuel raised without air (rev A, feed increase) - FT-005 reads high - F-100 fouling - hot ambient - FV-200 sticks	CO and smoke; outlet temperature falls although fuel rises; afterburning in convection; explosion risk	- AI-028 LO 1.0 % - AI-026 CO HI 200 ppm - operator - FY-005 ratio when added	4	3	12 H	R07, R08, R22, R25	Instrument · Operations · Process
N6.3	Flow	MORE	Excess air	- Air SP high - FT-005 reads low - high wind	Efficiency loss (more heat up the stack); lower flame temperature; possible flame instability	- AI-028 HI 6 % - AI-029 HI 7 %	2	3	6 M	R08	Instrument
N6.4	Temperature	MORE	Hot ambient air	Afternoon, summer	Less air mass through the same FV-200 opening; less stack effect; λ falls	- FIC-005 compensates while FV-200 has range - AI-028	3	3	9 M	R25	Process
N6.5	Pressure	LESS	Low windbox draft	- Damper closed - stack cold (start-up) - wind	Burner cannot draw air; air starvation	- PIAL-030 LO 1.5 - PSLL-030 trip 0.5 mmH₂O	4	2	8 M	R17	Mechanical
N6.6	Measurement	OTHER THAN	Air-flow meter wrong	FT-005 impulse line leak or drift	FIC-005 holds the wrong air; with a high reading the real air is cut → fuel-rich	- AI-028 O₂ independent check - rev B trim	4	2	8 M	R08, R22	Instrument · Operations
N6.7	Control	OTHER THAN	Trim fights tramp air	Rev B O ₂ trim with a peep door open (air leaks above the burner)	Trim reads high O ₂ and cuts burner air → CO at the burner	AI-026 CO HI	3	3	9 M	R08, R19	Instrument · Operations

Node N7 · Firebox (radiant section), burner and burner management system

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Node	N7 · Firebox (radiant section), burner and burner management system
Design intent	Burn the fuel completely in a stable flame at slightly negative pressure; transfer ~1.6 MW by radiation with bridgwall $\approx 990\text{ }^{\circ}\text{C}$; BMS purges, proves flame and trips on unsafe conditions.	Design conditions	$\approx 990\text{ }^{\circ}\text{C}$; $-2.5\text{ mmH}_2\text{O}$ at the arch; $\text{O}_2 \approx 2\%$
Equipment	Burner, flame scanner BS-005, refractory, explosion doors, peep doors, snuffing steam HV-221	Instruments	TIAH-023, PI-003, AI-028, BS-005, BMS
Drawing	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Document	NDFH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N7.1	Flame	NO	Flame failure while fuel flows	- Water in fuel - low fuel pressure - air starvation - scanner fault - snuffing steam opened	Unburnt fuel fills the firebox → explosion	- BSLL-005 trip (2 s) closes XV-301/302 - purge required before relight - explosion doors	5	1	5 M	R16, R20	Operations
N7.2	Temperature	MORE	Firebox too hot	- Over-firing - afterburning - tube rupture feeding fuel	Refractory and tube-support damage; high flux and tube skin	- TIAH-023 HI $1100\text{ }^{\circ}\text{C}$ - TIC-030 - TSHH-025	4	2	8 M	R21	Process
N7.3	Temperature	LESS	Firebox cold while firing	Air-rich or low firing	Unstable flame at low fire	- Burner turndown limit - BSLL-005	2	2	4 L	—	—
N7.4	Pressure	MORE	Positive firebox pressure	- Damper closed - high firing rate - puff on delayed ignition - tube rupture	Hot flue gas and flame out of peep doors and leaks: burns, structural damage	- PI-003 HI $-0.5\text{ mmH}_2\text{O}$ alarm - explosion doors - FO damper	4	2	8 M	R17, R18	Mechanical · Process / Instrument
N7.5	Composition	MORE	Too much combustibles (CO)	- Air shortage - poor atomisation	Afterburning, soot, explosion risk	- AI-026 CO HI - AI-028 LO	4	3	12 H	R07, R08, R22	Instrument · Operations
N7.6	Composition	AS WELL AS	Tramp air	- Peep door open - casing leaks	Wrong O_2 reading, efficiency loss, trim error	- AI-028 vs AI-029 comparison - rounds	2	3	6 M	R19	Operations
N7.7	Composition	AS WELL AS	Hydrocarbon from a leaking tube	Tube rupture	Uncontrolled fire in the box, high temperature, smoke	- TIAH-023 - PI-003 - FSLL-008 - HV-221 snuffing steam - ESD	5	2	10 H	R12, R20	Operations · Process
N7.8	Utility	OTHER THAN	Snuffing steam opened while firing	- Operator error - valve passing	Flame snuffed → fuel accumulation; wet steam damages refractory	- BSLL-005 trip - HV-221 normally closed	4	2	8 M	R20	Operations
N7.9	Safety system	OTHER THAN	BMS trip bypassed	Bypass left in after testing	Unsafe condition goes unprotected	Administrative control	5	2	10 H	R23	Operations

Node N8 · Convection section, stack and damper HV-400

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Node	N8 · Convection section, stack and damper HV-400
Design intent	Cool flue gas from ≈ 990 to 400 °C over the convection tubes and discharge it; produce ≈ -2.5 mmH ₂ O at the arch with damper HV-400 at 67 % (fail open).	Design conditions	Stack 400 °C; O ₂ ≈ 3.2 %; CO ≈ 20 ppm
Equipment	Convection bank, stack, damper HV-400	Instruments	TI-021, PI-031, PI-032, AI-026, AI-029, HIC-400
Drawing	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Document	NDFH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N8.1	Flow	LESS	Low draft / flue flow	- Damper closed or stuck - stack cold - convection fouled	Positive firebox pressure; air starvation	- PI-003 HI - PIAL-030 LO - PSLL-030 trip - FO damper	4	2	8 M	R17, R18	Mechanical · Process / Instrument
N8.2	Flow	MORE	Excess draft	- Damper wide open - high wind	Tramp air, efficiency loss, draft swings	- AI-029 HI - operator adjusts HV-400	2	3	6 M	—	—
N8.3	Temperature	MORE	Stack too hot	- Convection fouling (soot) - afterburning - over-firing	Efficiency loss; stack and damper overheating; indicates CO burning	- TI-021 HI 470 °C - AI-026	3	3	9 M	R22	Operations
N8.4	Temperature	LESS	Stack too cold	- Low firing - excess air	Acid dew-point corrosion with sulfur fuel	None (no low alarm)	2	2	4 L	R26	Instrument
N8.5	Composition	MORE	High CO at the stack	- Air shortage - poor atomisation	Toxic emission, afterburning, explosion risk in the convection section	AI-026 CO HI 200 ppm	4	3	12 H	R08, R22	Instrument · Operations
N8.6	Composition	MORE	High O ₂ at the stack	- Casing leaks - excess air	Efficiency loss; misleading control if the stack O ₂ is used	- AI-029 HI - control uses firebox AI-028	2	3	6 M	—	—

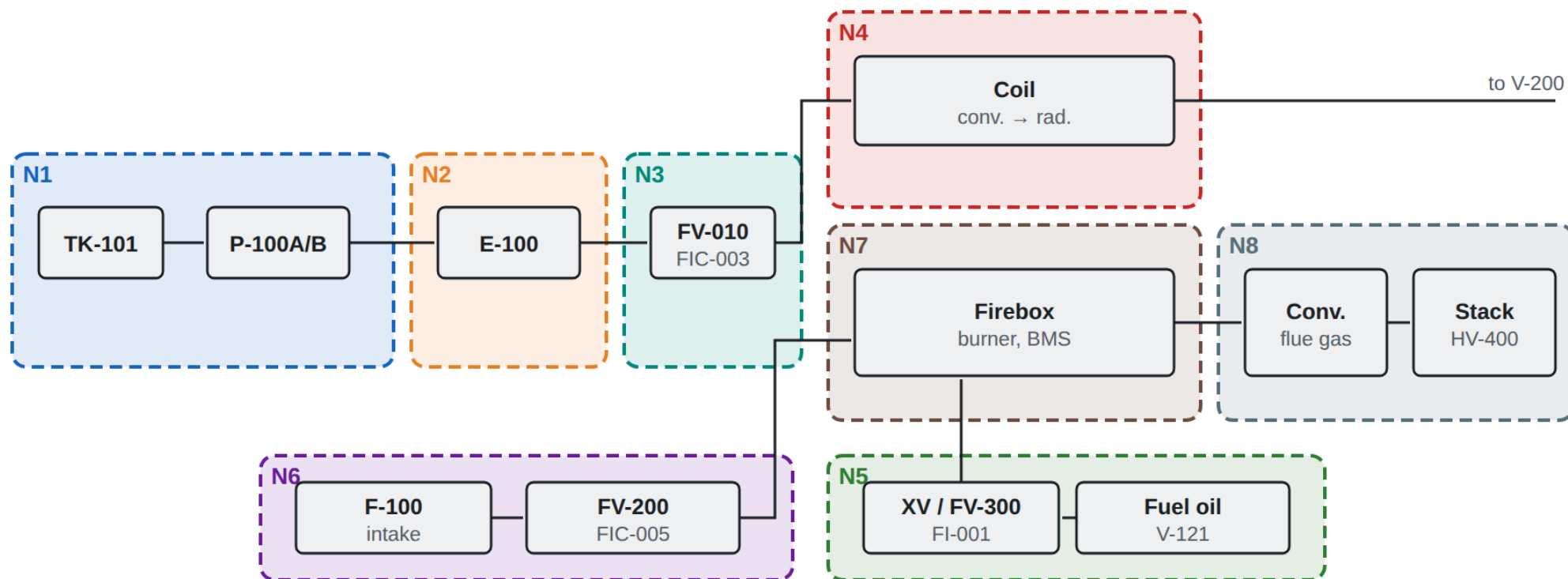
Node N9 · Operating procedures: start-up (purge, light-off) and shutdown

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Node	N9 · Operating procedures: start-up (purge, light-off) and shutdown
Design intent	Start: feed flow established, draft set, 5-min purge (air $\geq$ 1700 kg/h, damper $\geq$ 60 %), light-off at 10–20 % fuel with a 10-s trial, then ramp to 350 °C. Shutdown: cut fuel, burner out, keep feed until cool.	Design conditions	Procedure steps in the Exercises tab
Equipment	BMS sequence, operator actions	Instruments	BMS screen, FIC-003, FIC-005, TIC-030 (MAN), PI-004
Drawing	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Document	NDFH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N9.1	Purge	NO	Light-off without purge	- Purge bypassed - BMS fault	Explosion of accumulated fuel vapour	- BMS purge permissives and timer - explosion doors	5	1	5 M	R15, R23	Instrument · Operations
N9.2	Purge	LESS	Purge too short or at low air	- Air below 1700 kg/h - damper closed	Incomplete removal of combustibles	Purge permissives (air, damper, draft) stop the timer	4	1	4 L	R23	Operations
N9.3	Light-off	MORE	Too much fuel at light-off	TIC-030 not in MAN or output above 20 %	Delayed ignition, puff, explosion doors lift	BMS low-fire permissive (10–20 % MAN)	4	1	4 L	R27	Operations
N9.4	Light-off	LATE	Delayed ignition	- Igniter fault - poor atomisation	Fuel accumulates during the trial → puff	10-s trial for ignition then trip - purge before retry	4	2	8 M	R15, R27	Instrument · Operations
N9.5	Sequence	BEFORE	Fuel before feed flow	Step order not followed	Tubes fired dry → overheating	FSLL-008 as purge permissive and trip	4	1	4 L	R27	Operations
N9.6	Sequence	BEFORE	Fuel raised before air (load increase)	Operator raises fuel/feed first	Fuel-rich firing, CO	- AI-028, AI-026 alarms - procedure "air leads on increase"	4	3	12 H	R07, R22	Instrument · Operations
N9.7	Sequence	AFTER	Feed stopped before the burner is out (shutdown)	Wrong order	Tubes overheated	FSLL-008 trip	4	1	4 L	R27	Operations
N9.8	Relight	EARLY	Relight straight after a flame failure	Operator tries to relight at once	Explosion of fuel left in the box	BMS forces trip reset and full purge	5	1	5 M	R23	Operations
N9.9	Shutdown	OTHER THAN	Isolation incomplete for maintenance	XV passing, V-121 not closed, bleed not open	Fuel enters the firebox during entry	- Double block and bleed - permit to work	5	1	5 M	R15	Instrument

Appendix 2 · HAZOP documents (drawings with node boundaries)

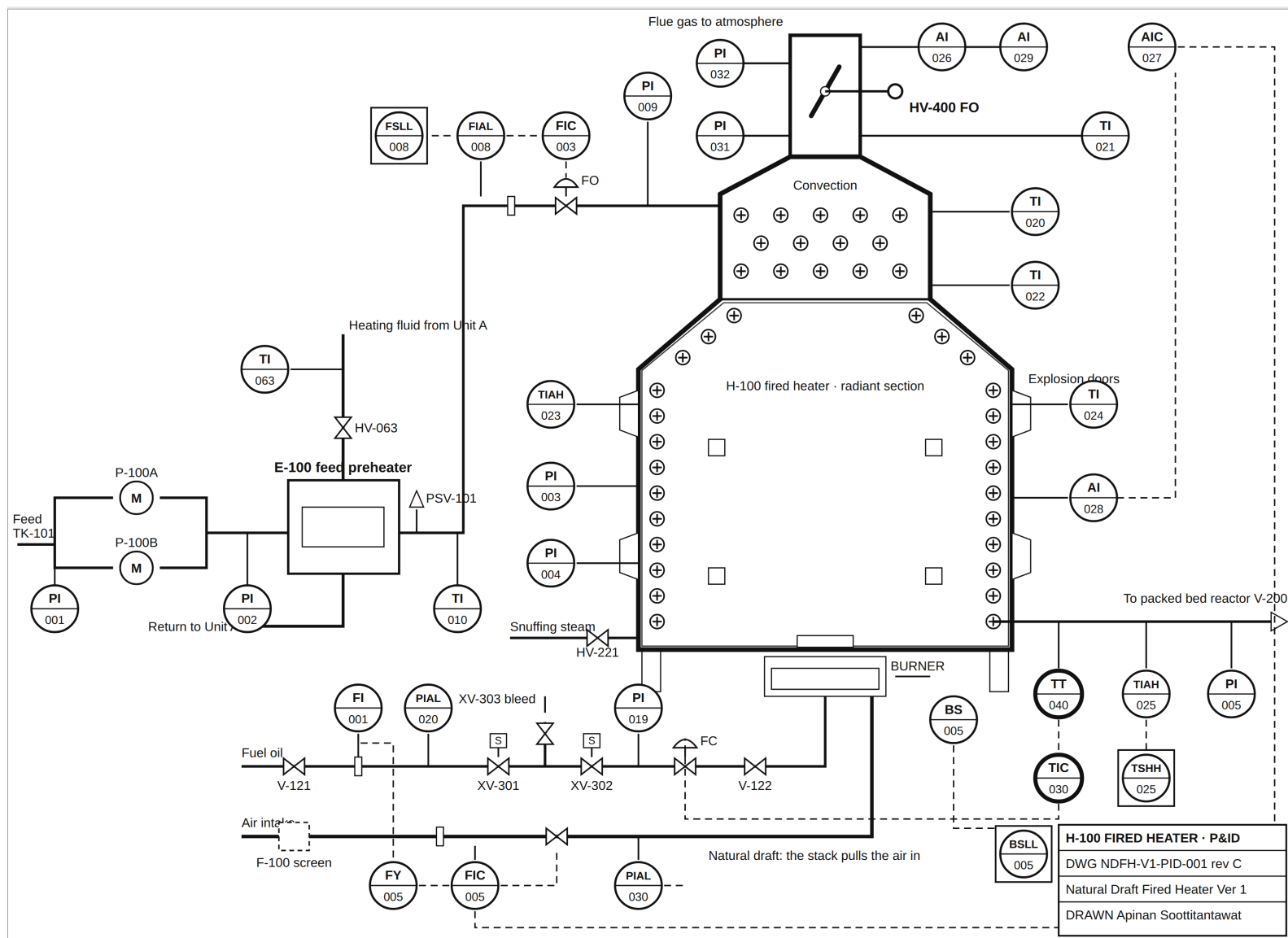
A2-1 · HAZOP node map



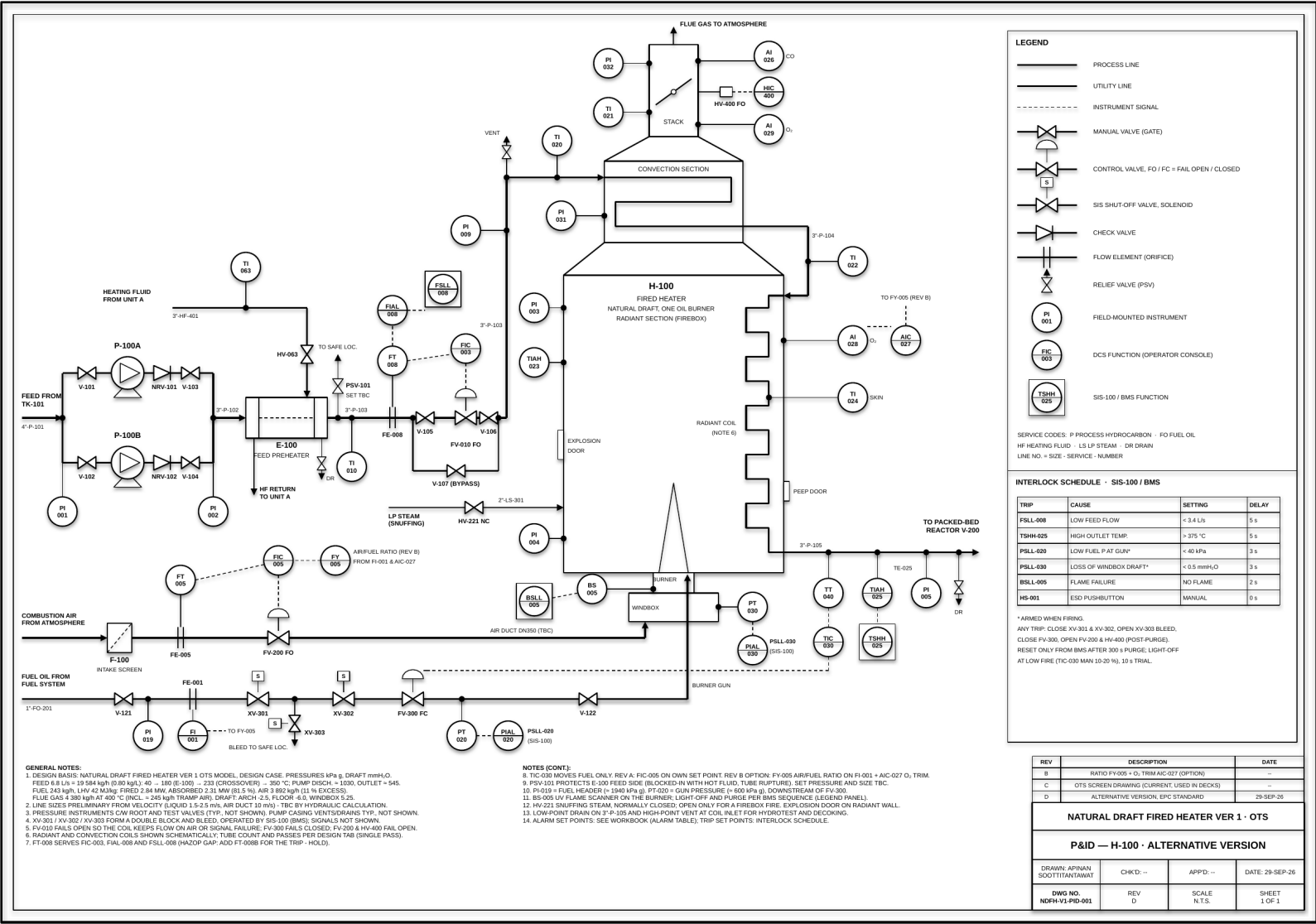
N9 (not drawn): start-up, shutdown and purge procedures

A2-1. HAZOP node map of H-100 (nodes N1–N8; N9 = procedures).

P&ID of H-100, rev C, as drawn on the P&ID tab of the simulator.



A2-3 · NDFH-V1-PID-001 rev D (alternative EPC drawing)



P&ID of H-100, rev D, EPC-standard alternative version with interlock schedule.

Appendix 3 · HAZOP action close-out sheets

One sheet per recommendation (27 sheets). The action party writes the response, ticks the status and signs; the HAZOP leader accepts the close-out.

HAZOP ACTION CLOSE-OUT SHEET · R01

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R01
Node	N1 · Feed supply: TK-101 → P-100A/B → E-100 inlet	Action party	Instrument / Operations
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Flow NO : No feed flow to E-100	- Running pump P-100A trips (motor, power) - suction valve closed - TK-101 empty - P-100B fails to start	Coil loses its coolant while firing → tube overheating, coking, rupture and fire in the firebox	- FSL-008 trips the burner (< 3.4 L/s, 5 s) - FIAL-008 LO alarm - P-100B spare (manual start)	S4 L2 8 Medium

HAZOP recommendation

Auto-start spare P-100B on low discharge pressure (PI-002) or P-100A trip, with the start completing inside the FSL-008 delay; test the spare monthly.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R02

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R02
Node	N1 · Feed supply: TK-101 → P-100A/B → E-100 inlet	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Flow NO : No feed flow to E-100	- Running pump P-100A trips (motor, power) - suction valve closed - TK-101 empty - P-100B fails to start	Coil loses its coolant while firing → tube overheating, coking, rupture and fire in the firebox	- FSSL-008 trips the burner (< 3.4 L/s, 5 s) - FIAL-008 LO alarm - P-100B spare (manual start)	S4 L2 8 Medium
N1.2	Flow LESS : Less feed flow	- Pump wear or cavitation - partly closed suction valve - suction strainer plugged - low TK-101 level	Higher coil outlet temperature and skin temperature at the same firing; production shortfall	- TIC-030 cuts fuel - FIAL-008 LO - FSSL-008 at 3.4 L/s	S3 L3 9 Medium
N1.6	Pressure LESS : Low suction pressure	- Low TK-101 level - plugged suction strainer - hot feed near bubble point	Cavitation, pump damage, flow loss → see NO flow	- PI-001 (local) - operator rounds	S3 L3 9 Medium
N1.7	Level LESS : Low level in TK-101	- Supply from upstream stopped - tank drained	Pumps lose suction → no flow	Operator monitoring only (no level instrument on the P&ID)	S3 L3 9 Medium
N1.8	Level MORE : High level in TK-101	Upstream supply more than consumption	Tank overflow, spill, fire risk (outside this node)	Tank overflow line (upstream system)	S3 L2 6 Medium

HAZOP recommendation

Add level transmitter LT-101 on TK-101 with low-level alarm LAL and low-low pump trip LSL to protect P-100A/B from running dry.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R03

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R03
Node	N1 · Feed supply: TK-101 → P-100A/B → E-100 inlet	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.5	Pressure MORE : High discharge pressure	- FV-010 or downstream valve closed (deadhead) - V-200 blocked	Pump overheats at no flow; possible seal failure and leak (fire)	- Pump shut-off head below piping design pressure - PI-002	S3 L2 6 Medium

HAZOP recommendation

Provide minimum-flow recirculation (restriction orifice back to TK-101) on P-100A/B so a closed FV-010 cannot deadhead the running pump.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R04

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R04
Node	N1 · Feed supply: TK-101 → P-100A/B → E-100 inlet	Action party	Mechanical
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.4	Flow REVERSE : Reverse flow through the idle pump	Passing check valve on P-100B discharge	Recirculation through the idle pump, less flow forward, idle pump spins backwards	- Check valves - FIAL-008 LO	S2 L2 4 Low
N3.4	Flow REVERSE : Back-flow from the coil	Pump stops with coil at higher pressure (V-200 pressure)	Hot hydrocarbon returns toward E-100 and pumps	Pump discharge check valves	S2 L2 4 Low

HAZOP recommendation

Confirm a check valve on each pump discharge; include in the inspection plan (passing check valve spins the idle pump backwards).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R05

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R05
Node	N2 · E-100 feed preheater (feed and hot-fluid sides)	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.4	Pressure MORE : Feed side blocked in with hot fluid flowing	Inlet and outlet feed valves closed during an outage	Thermal expansion / vaporisation → overpressure of the E-100 feed side	PSV-101	S4 L2 8 Medium
N2.5	Integrity OTHER THAN : E-100 tube leak	Corrosion, vibration	Hot fluid and feed mix; contamination of either system; high-pressure side flows into the low-pressure side	• PSV-101 (if feed is the low-pressure side) • product quality checks	S3 L2 6 Medium

HAZOP recommendation

Verify PSV-101 is sized for blocked-in feed with hot fluid flowing (thermal expansion and vaporisation) and for E-100 tube rupture (two-thirds rule).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R06

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R06
Node	N2 · E-100 feed preheater (feed and hot-fluid sides)	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.1	Temperature LESS : Feed to H-100 colder than 180 °C	- HV-063 closed or Unit A cuts hot fluid - E-100 fouling - hot fluid colder	Heater must fire harder: more fuel, higher flux and tube skin; outlet may droop if air does not follow	- TI-010 LO alarm 170 °C - TIC-030 adds fuel - TI-024 HI 430 °C	S3 L3 9 Medium
N2.3	Flow (hot fluid) NO : No hot fluid	- Unit A trip - HV-063 closed	Feed enters at about 40 °C: firing needed nearly doubles, beyond air/draft capacity; outlet falls; possible fuel-rich firing	- TI-010 LO alarm - TIC-030 - AI-028, AI-026 alarms	S3 L3 9 Medium
N2.6	Heat transfer LESS : Fouling	Deposits on the feed or hot side	Gradual fall in TI-010 and rise in firing; efficiency loss	- TI-010 LO alarm - cleaning programme	S2 L4 8 Medium

HAZOP recommendation

Add a hot-fluid return temperature indicator and an E-100 approach-temperature calculation so fouling is seen before TI-010 falls to its alarm.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R07

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R07
Node	N5/N6 · Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun / Combustion air: F-100 intake screen → FV-200 → windbox and burner register	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.3	Flow MORE : More feed flow	- Operator raises FIC-003 SP - FV-010 fails open (FO)	More duty required; fuel rises; air may not follow (rev A) → fuel-rich firebox, CO, smoke	- TIC-030 - AI-028 O₂ LO alarm - AI-026 CO HI alarm - operator raises air first	S3 L3 9 Medium
N2.1	Temperature LESS : Feed to H-100 colder than 180 °C	- HV-063 closed or Unit A cuts hot fluid - E-100 fouling - hot fluid colder	Heater must fire harder: more fuel, higher flux and tube skin; outlet may droop if air does not follow	- TI-010 LO alarm 170 °C - TIC-030 adds fuel - TI-024 HI 430 °C	S3 L3 9 Medium
N2.3	Flow (hot fluid) NO : No hot fluid	- Unit A trip - HV-063 closed	Feed enters at about 40 °C: firing needed nearly doubles, beyond air/draft capacity; outlet falls; possible fuel-rich firing	- TI-010 LO alarm - TIC-030 - AI-028, AI-026 alarms	S3 L3 9 Medium
N3.3	Flow MORE : High flow	- FT-003 reads low - FV-010 fails open	Duty above capacity; outlet temperature falls; fuel-rich firing if air lags	- TIC-030 - AI-028/AI-026 alarms	S3 L2 6 Medium
N4.2	Temperature LESS : Outlet below 350 °C	- Air-limited (fuel-rich) firing - fuel supply low - nozzle plugged - feed colder or higher	Off-spec reaction in V-200; operator may add fuel into a fuel-rich box	- TIC-030 LO 344 °C - AI-028 O₂ LO - AI-026 CO HI	S3 L3 9 Medium
N5.3	Flow MORE : More fuel than needed	- TIC-030 fault - TT-040 reads low - operator in MAN - FV-300 sticks open	High outlet temperature; if air does not follow → fuel-rich firing, CO, afterburning, explosion risk	- TIC-030 HI - TSHH-025 - AI-028 LO - AI-026 CO HI - FV-300 FC	S4 L2 8 Medium
N6.2	Flow LESS : Less air than needed ($\lambda \rightarrow 1$ or below)	- Fuel raised without air (rev A, feed increase) - FT-005 reads high - F-100 fouling - hot ambient - FV-200 sticks	CO and smoke; outlet temperature falls although fuel rises; afterburning in convection; explosion risk	- AI-028 LO 1.0 % - AI-026 CO HI 200 ppm - operator - FY-005 ratio when added	S4 L3 12 High
N7.5	Composition MORE : Too much combustibles (CO)	- Air shortage - poor atomisation	Afterburning, soot, explosion risk	- AI-026 CO HI - AI-028 LO	S4 L3 12 High
N9.6	Sequence BEFORE : Fuel raised before air (load increase)	Operator raises fuel/feed first	Fuel-rich firing, CO	- AI-028, AI-026 alarms - procedure "air leads on increase"	S4 L3 12 High

HAZOP recommendation

Implement cross-limiting (lead-lag) air/fuel control: fuel demand limited by measured air, air demand not below that needed for measured fuel.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R08

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R08
Node	N6 · Combustion air: F-100 intake screen → FV-200 → windbox and burner register	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.3	Flow MORE : More fuel than needed	- TIC-030 fault - TT-040 reads low - operator in MAN - FV-300 sticks open	High outlet temperature; if air does not follow → fuel-rich firing, CO, afterburning, explosion risk	- TIC-030 HI - TSHH-025 - AI-028 LO - AI-026 CO HI - FV-300 FC	S4 L2 8 Medium
N5.7	Composition OTHER THAN : Different fuel quality	- New fuel batch with different LHV - water in fuel	Changed air requirement; sputtering flame, flame-out (water)	- AI-028 O₂ - BSLL-005 flame failure trip	S3 L3 9 Medium
N5.8	Measurement OTHER THAN : FI-001 reads wrong	- Wrong orifice plate - drift	Ratio FY-005 (if used) sets wrong air; wrong efficiency calculations	AI-028 O ₂ as independent check	S3 L2 6 Medium
N6.2	Flow LESS : Less air than needed ($\lambda \rightarrow 1$ or below)	- Fuel raised without air (rev A, feed increase) - FT-005 reads high - F-100 fouling - hot ambient - FV-200 sticks	CO and smoke; outlet temperature falls although fuel rises; afterburning in convection; explosion risk	- AI-028 LO 1.0 % - AI-026 CO HI 200 ppm - operator - FY-005 ratio when added	S4 L3 12 High
N6.3	Flow MORE : Excess air	- Air SP high - FT-005 reads low - high wind	Efficiency loss (more heat up the stack); lower flame temperature; possible flame instability	- AI-028 HI 6 % - AI-029 HI 7 %	S2 L3 6 Medium
N6.6	Measurement OTHER THAN : Air-flow meter wrong	FT-005 impulse line leak or drift	FIC-005 holds the wrong air; with a high reading the real air is cut → fuel-rich	- AI-028 O₂ independent check - rev B trim	S4 L2 8 Medium
N6.7	Control OTHER THAN : Trim fights tramp air	Rev B O ₂ trim with a peep door open (air leaks above the burner)	Trim reads high O ₂ and cuts burner air → CO at the burner	AI-026 CO HI	S3 L3 9 Medium
N7.5	Composition MORE : Too much combustibles (CO)	- Air shortage - poor atomisation	Afterburning, soot, explosion risk	- AI-026 CO HI - AI-028 LO	S4 L3 12 High
N8.5	Composition MORE : High CO at the stack	- Air shortage - poor atomisation	Toxic emission, afterburning, explosion risk in the convection section	AI-026 CO HI 200 ppm	S4 L3 12 High

HAZOP recommendation

Put the O₂ trim (rev B) in service with a CO override: high CO blocks the trim from cutting air further.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

--

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			

Sign-off	Name	Signature	Date
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R09

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R09
Node	N5 · Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.2	Flow LESS : Less fuel than demanded	- Supply pressure falls - nozzle partly plugged - strainer fouling - FV-300 sticks	Outlet temperature falls; FV-300 opens to 100 %; low gun pressure → poor atomisation, unstable flame	- PIAL-020 LO 100 kPa - PSLL-020 trip 40 kPa - TIC-030 LO alarm	S3 L3 9 Medium

HAZOP recommendation

Cascade TIC-030 to a fuel-flow controller on FI-001 so fuel-pressure disturbances are corrected before they reach the outlet temperature.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R10

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R10
Node	N4 · H-100 process coil: convection → crossover → radiant → outlet to V-200	Action party	Process / Instrument
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.2	Flow LESS : Low flow	- FIC-003 SP lowered too far - FT reads high	High skin and outlet temperatures; coking	- FIAL-008 LO 5.0 L/s - FSLL-008 3.4 L/s - TIC-030	S3 L3 9 Medium
N4.3	Temperature (tube metal) MORE : Tube skin above limit	- Coke inside tubes - low flow - flame impingement - over-firing	Creep damage, bulging, tube rupture → fire	- TI-024 HI 430 °C - FSLL-008 - TIC-030	S4 L3 12 High
N4.7	Composition AS WELL AS : Coke builds up	High skin temperature, low velocity, off-spec feed	Rising skin temperature and pressure drop, lower heat pickup	- TI-024 trend - decoking programme	S3 L3 9 Medium

HAZOP recommendation

Add more radiant tube-skin thermocouples (hottest tubes) and evaluate a TI-024 high-high trip by LOPA.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R11

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R11
Node	N4 · H-100 process coil: convection → crossover → radiant → outlet to V-200	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.5	Pressure MORE : High discharge pressure	- FV-010 or downstream valve closed (deadhead) - V-200 blocked	Pump overheats at no flow; possible seal failure and leak (fire)	- Pump shut-off head below piping design pressure - PI-002	S3 L2 6 Medium
N4.4	Pressure MORE : Coil overpressure	Outlet blocked (V-200 inlet valve closed) with pump running	Coil rupture if pump shut-off exceeds design pressure; fire	PI-005 indication only	S5 L1 5 Medium

HAZOP recommendation

Confirm coil design pressure $\geq$ pump shut-off head; otherwise add overpressure protection. Add high-pressure alarm on PI-005 (blocked outlet to V-200).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

--

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R12

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R12
Node	N4 · H-100 process coil: convection → crossover → radiant → outlet to V-200	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Flow NO : No flow to the coil	- FV-010 closed by controller fault or operator - line blocked	Tube overheating while firing → coking, rupture, firebox fire	- FSL-008 trip (independent element, 5 s) - FIAL-008 LO - FV-010 fails open on air failure	S4 L2 8 Medium
N4.5	Pressure LESS : Loss of containment	Tube rupture (corrosion, creep, overheating)	Hydrocarbon burns in the firebox: firebox temperature and pressure rise, smoke, possible explosion	- TIAH-023 HI - PI-003 HI - FSL-008 as flow drops - snuffing steam HV-221 - ESD	S5 L2 10 High
N5.9	Integrity OTHER THAN : Fuel leak at the burner front	Gasket or hose failure	Pool fire at the heater floor	- Fire detection and fire fighting - V-121 isolation - ESD	S4 L2 8 Medium
N7.7	Composition AS WELL AS : Hydrocarbon from a leaking tube	Tube rupture	Uncontrolled fire in the box, high temperature, smoke	- TIAH-023 - PI-003 - FSL-008 - HV-221 snuffing steam - ESD	S5 L2 10 High

HAZOP recommendation

Provide remotely operated emergency isolation valves on the heater inlet and outlet, operable from the control room, for tube rupture.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R13

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R13
Node	N5 · Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.6	Temperature LESS : Cold, viscous fuel oil	• Tracing or heater failure • cold weather	Poor atomisation, long smoky flame, CO, soot on convection tubes	• Operator observation • AI-026 CO HI	S3 L3 9 Medium

HAZOP recommendation

Add fuel-oil temperature indication and low-temperature alarm; keep oil at atomising viscosity (tracing or heater).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R14

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R14
Node	N5 · Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	Action party	Mechanical
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.1	Flow NO : No fuel to the burner	- Supply pump trip - V-121 closed - XV closed - strainer blocked	Flame-out; outlet temperature falls; risk of relighting without purge	- PSLL-020 and BSLL-005 trips - BMS requires purge before relight	S3 L3 9 Medium
N5.2	Flow LESS : Less fuel than demanded	- Supply pressure falls - nozzle partly plugged - strainer fouling - FV-300 sticks	Outlet temperature falls; FV-300 opens to 100 %; low gun pressure → poor atomisation, unstable flame	- PIAL-020 LO 100 kPa - PSLL-020 trip 40 kPa - TIC-030 LO alarm	S3 L3 9 Medium
N5.5	Pressure MORE : High fuel pressure at the gun	- Control valve open fully - plugged gun tip (pressure rises for same flow)	Flame lift-off / instability; high firing	PIAL-020 HI 1400 kPa	S3 L2 6 Medium

HAZOP recommendationInstall a duplex strainer with ΔP indication upstream of FV-300.**Response of the action party** (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R15

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R15
Node	N5 · Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.4	Flow AS WELL AS : Fuel leaks into the firebox while shut down	• XV-301/302 passing • bleed XV-303 blocked	Fuel accumulates; explosion at the next light-off	• Double block and bleed • BMS purge (5 min) before light-off	S5 L2 10 High
N9.1	Purge NO : Light-off without purge	• Purge bypassed • BMS fault	Explosion of accumulated fuel vapour	• BMS purge permissives and timer • explosion doors	S5 L1 5 Medium
N9.4	Light-off LATE : Delayed ignition	• Igniter fault • poor atomisation	Fuel accumulates during the trial → puff	• 10-s trial for ignition then trip • purge before retry	S4 L2 8 Medium
N9.9	Shutdown OTHER THAN : Isolation incomplete for maintenance	XV passing, V-121 not closed, bleed not open	Fuel enters the firebox during entry	• Double block and bleed • permit to work	S5 L1 5 Medium

HAZOP recommendation

Add proof-of-closure switches on XV-301/302 as purge permissives and a valve-leak test before every light-off.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R16

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R16
Node	N5 · Fuel oil train: V-121 → FI-001 → XV-301/303/302 → FV-300 → V-122 → burner gun	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.9	Composition AS WELL AS : Water in the feed	Tank bottoms water carried over	Water flashes in the coil: pressure surges, hammer, flow upset, TIC-030 disturbance	None specific	S3 L2 6 Medium
N5.7	Composition OTHER THAN : Different fuel quality	- New fuel batch with different LHV - water in fuel	Changed air requirement; sputtering flame, flame-out (water)	- AI-028 O₂ - BSLL-005 flame failure trip	S3 L3 9 Medium
N7.1	Flame NO : Flame failure while fuel flows	- Water in fuel - low fuel pressure - air starvation - scanner fault - snuffing steam opened	Unburnt fuel fills the firebox → explosion	- BSLL-005 trip (2 s) closes XV-301/302 - purge required before relight - explosion doors	S5 L1 5 Medium

HAZOP recommendation

Water-draw and sampling procedure for the fuel-oil tank; alarm on water in fuel if available.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)
Status
☐ Accepted and done

☐ Accepted with alternative

☐ Rejected

☒ Open (at issue)

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R17

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R17
Node	N8 · Convection section, stack and damper HV-400	Action party	Mechanical
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.1	Flow NO : No combustion air	- FV-200 closed by fault - windbox draft lost (damper closed) - F-100 blocked	Flame goes fuel-rich then out; unburnt fuel in the box → explosion on re-ignition	- PSLL-030 trip (< 0.5 mmH₂O draft, 3 s) - BSLL-005 - FV-200 fails open	S5 L1 5 Medium
N6.5	Pressure LESS : Low windbox draft	- Damper closed - stack cold (start-up) - wind	Burner cannot draw air; air starvation	- PIAL-030 LO 1.5 - PSLL-030 trip 0.5 mmH₂O	S4 L2 8 Medium
N7.4	Pressure MORE : Positive firebox pressure	- Damper closed - high firing rate - puff on delayed ignition - tube rupture	Hot flue gas and flame out of peep doors and leaks: burns, structural damage	- PI-003 HI –0.5 mmH₂O alarm - explosion doors - FO damper	S4 L2 8 Medium
N8.1	Flow LESS : Low draft / flue flow	- Damper closed or stuck - stack cold - convection fouled	Positive firebox pressure; air starvation	- PI-003 HI - PIAL-030 LO - PSLL-030 trip - FO damper	S4 L2 8 Medium

HAZOP recommendation

Fit a mechanical minimum-open stop on damper HV-400 (e.g. 30 %) so it cannot close fully.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R18

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R18
Node	N7/N8 · Firebox (radiant section), burner and burner management system / Convection section, stack and damper HV-400	Action party	Process / Instrument
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.1	Flow NO : No combustion air	- FV-200 closed by fault - windbox draft lost (damper closed) - F-100 blocked	Flame goes fuel-rich then out; unburnt fuel in the box → explosion on re-ignition	- PSLL-030 trip (< 0.5 mmH₂O draft, 3 s) - BSLL-005 - FV-200 fails open	S5 L1 5 Medium
N7.4	Pressure MORE : Positive firebox pressure	- Damper closed - high firing rate - puff on delayed ignition - tube rupture	Hot flue gas and flame out of peep doors and leaks: burns, structural damage	- PI-003 HI –0.5 mmH₂O alarm - explosion doors - FO damper	S4 L2 8 Medium
N8.1	Flow LESS : Low draft / flue flow	- Damper closed or stuck - stack cold - convection fouled	Positive firebox pressure; air starvation	- PI-003 HI - PIAL-030 LO - PSLL-030 trip - FO damper	S4 L2 8 Medium

HAZOP recommendation

Evaluate a high firebox-pressure trip (PSHH on PI-003) by LOPA; today PI-003 only alarms.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R19

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R19
Node	N7 · Firebox (radiant section), burner and burner management system	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.7	Control OTHER THAN : Trim fights tramp air	Rev B O ₂ trim with a peep door open (air leaks above the burner)	Trim reads high O ₂ and cuts burner air → CO at the burner	AI-026 CO HI	S3 L3 9 Medium
N7.6	Composition AS WELL AS : Tramp air	- Peep door open - casing leaks	Wrong O ₂ reading, efficiency loss, trim error	- AI-028 vs AI-029 comparison - rounds	S2 L3 6 Medium

HAZOP recommendation

Add peep-door and explosion-door checks to the field operator round sheet; peep doors closed except when viewing.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R20

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R20
Node	N7 · Firebox (radiant section), burner and burner management system	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.1	Flame NO : Flame failure while fuel flows	- Water in fuel - low fuel pressure - air starvation - scanner fault - snuffing steam opened	Unburnt fuel fills the firebox → explosion	- BSLL-005 trip (2 s) closes XV-301/302 - purge required before relight - explosion doors	S5 L1 5 Medium
N7.7	Composition AS WELL AS : Hydrocarbon from a leaking tube	Tube rupture	Uncontrolled fire in the box, high temperature, smoke	- TIAH-023 - PI-003 - FSL-008 - HV-221 snuffing steam - ESD	S5 L2 10 High
N7.8	Utility OTHER THAN : Snuffing steam opened while firing	- Operator error - valve passing	Flame snuffed → fuel accumulation; wet steam damages refractory	- BSLL-005 trip - HV-221 normally closed	S4 L2 8 Medium

HAZOP recommendation

Lock HV-221 snuffing steam closed while firing; drain condensate before use; write it into the fire-emergency procedure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R21

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R21
Node	N7 · Firebox (radiant section), burner and burner management system	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.5	Pressure LESS : Loss of containment	Tube rupture (corrosion, creep, overheating)	Hydrocarbon burns in the firebox: firebox temperature and pressure rise, smoke, possible explosion	• TIAH-023 HI • PI-003 HI • FSLL-008 as flow drops • snuffing steam HV-221 • ESD	S5 L2 10 High
N7.2	Temperature MORE : Firebox too hot	• Over-firing • afterburning • tube rupture feeding fuel	Refractory and tube-support damage; high flux and tube skin	• TIAH-023 HI 1100 °C • TIC-030 • TSHH-025	S4 L2 8 Medium

HAZOP recommendation

Evaluate a high firebox-temperature trip (TSHH on TIAH-023) by LOPA.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R22

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R22
Node	N6/N7 · Combustion air: F-100 intake screen → FV-200 → windbox and burner register / Firebox (radiant section), burner and burner management system	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.3	Flow MORE : More feed flow	- Operator raises FIC-003 SP - FV-010 fails open (FO)	More duty required; fuel rises; air may not follow (rev A) → fuel-rich firebox, CO, smoke	- TIC-030 - AI-028 O₂ LO alarm - AI-026 CO HI alarm - operator raises air first	S3 L3 9 Medium
N2.3	Flow (hot fluid) NO : No hot fluid	- Unit A trip - HV-063 closed	Feed enters at about 40 °C: firing needed nearly doubles, beyond air/draft capacity; outlet falls; possible fuel-rich firing	- TI-010 LO alarm - TIC-030 - AI-028, AI-026 alarms	S3 L3 9 Medium
N4.2	Temperature LESS : Outlet below 350 °C	- Air-limited (fuel-rich) firing - fuel supply low - nozzle plugged - feed colder or higher	Off-spec reaction in V-200; operator may add fuel into a fuel-rich box	- TIC-030 LO 344 °C - AI-028 O₂ LO - AI-026 CO HI	S3 L3 9 Medium
N5.3	Flow MORE : More fuel than needed	- TIC-030 fault - TT-040 reads low - operator in MAN - FV-300 sticks open	High outlet temperature; if air does not follow → fuel-rich firing, CO, afterburning, explosion risk	- TIC-030 HI - TSHH-025 - AI-028 LO - AI-026 CO HI - FV-300 FC	S4 L2 8 Medium
N6.2	Flow LESS : Less air than needed (λ → 1 or below)	- Fuel raised without air (rev A, feed increase) - FT-005 reads high - F-100 fouling - hot ambient - FV-200 sticks	CO and smoke; outlet temperature falls although fuel rises; afterburning in convection; explosion risk	- AI-028 LO 1.0 % - AI-026 CO HI 200 ppm - operator - FY-005 ratio when added	S4 L3 12 High
N6.6	Measurement OTHER THAN : Air-flow meter wrong	FT-005 impulse line leak or drift	FIC-005 holds the wrong air; with a high reading the real air is cut → fuel-rich	- AI-028 O₂ independent check - rev B trim	S4 L2 8 Medium
N7.5	Composition MORE : Too much combustibles (CO)	- Air shortage - poor atomisation	Afterburning, soot, explosion risk	- AI-026 CO HI - AI-028 LO	S4 L3 12 High
N8.3	Temperature MORE : Stack too hot	- Convection fouling (soot) - afterburning - over-firing	Efficiency loss; stack and damper overheating; indicates CO burning	- TI-021 HI 470 °C - AI-026	S3 L3 9 Medium
N8.5	Composition MORE : High CO at the stack	- Air shortage - poor atomisation	Toxic emission, afterburning, explosion risk in the convection section	AI-026 CO HI 200 ppm	S4 L3 12 High
N9.6	Sequence BEFORE : Fuel raised before air (load increase)	Operator raises fuel/feed first	Fuel-rich firing, CO	- AI-028, AI-026 alarms - procedure "air leads on increase"	S4 L3 12 High

HAZOP recommendation

Written response procedure for high CO (AI-026) and low O₂ (AI-028): raise air first, then cut fuel; train on the simulator.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

--

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R23

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R23
Node	N9 · Operating procedures: start-up (purge, light-off) and shutdown	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.9	Safety system OTHER THAN : BMS trip bypassed	Bypass left in after testing	Unsafe condition goes unprotected	Administrative control	S5 L2 10 High
N9.1	Purge NO : Light-off without purge	- Purge bypassed - BMS fault	Explosion of accumulated fuel vapour	- BMS purge permissives and timer - explosion doors	S5 L1 5 Medium
N9.2	Purge LESS : Purge too short or at low air	- Air below 1700 kg/h - damper closed	Incomplete removal of combustibles	Purge permissives (air, damper, draft) stop the timer	S4 L1 4 Low
N9.8	Relight EARLY : Relight straight after a flame failure	Operator tries to relight at once	Explosion of fuel left in the box	BMS forces trip reset and full purge	S5 L1 5 Medium

HAZOP recommendation

Trip-bypass management: any BMS bypass needs a permit, time limit and MOC; bypass status shown on the DCS.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R24

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R24
Node	N4 · H-100 process coil: convection → crossover → radiant → outlet to V-200	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.1	Temperature MORE : Outlet above 350 °C	- TIC-030 fault or in MAN with high output - TT-040 reads low - feed flow falls - set point error	Cracking/coking in the coil, catalyst damage in V-200, tube creep	- TIC-030 HI 356 °C - TIAH-025 HI 360 °C - TSHH-025 trip 375 °C (independent element)	S4 L2 8 Medium
N4.8	Measurement OTHER THAN : Outlet transmitter wrong	TT-040 drift or failure	TIC-030 over- or under-fires; TIAH-025 disagrees	Independent TIAH-025 / TSHH-025	S3 L3 9 Medium

HAZOP recommendation

Add a deviation alarm TT-040 vs TIAH-025 (outlet transmitter disagreement).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R25

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R25
Node	N6 · Combustion air: F-100 intake screen → FV-200 → windbox and burner register	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.2	Flow LESS : Less air than needed ($\lambda \rightarrow 1$ or below)	- Fuel raised without air (rev A, feed increase) - FT-005 reads high - F-100 fouling - hot ambient - FV-200 sticks	CO and smoke; outlet temperature falls although fuel rises; afterburning in convection; explosion risk	- AI-028 LO 1.0 % - AI-026 CO HI 200 ppm - operator - FY-005 ratio when added	S4 L3 12 High
N6.4	Temperature MORE : Hot ambient air	Afternoon, summer	Less air mass through the same FV-200 opening; less stack effect; λ falls	- FIC-005 compensates while FV-200 has range - AI-028	S3 L3 9 Medium

HAZOP recommendation

Check FV-200 and F-100 capacity at the hottest ambient and highest firing; add intake-screen ΔP indication.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R26

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R26
Node	N8 · Convection section, stack and damper HV-400	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.4	Temperature LESS : Stack too cold	• Low firing • excess air	Acid dew-point corrosion with sulfur fuel	None (no low alarm)	S2 L2 4 Low

HAZOP recommendation

Low stack-temperature alarm on TI-021 when firing sulfur-bearing oil (acid dew point).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R27

Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator	Document	NDFH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R27
Node	N9 · Operating procedures: start-up (purge, light-off) and shutdown	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.10	Maintenance OTHER THAN : Pump change-over done incorrectly	Spare started with discharge valve closed, or running pump stopped first	Flow interruption → FSLL-008 trip of the heater	• Procedure • FSLL-008	S2 L3 6 Medium
N9.3	Light-off MORE : Too much fuel at light-off	TIC-030 not in MAN or output above 20 %	Delayed ignition, puff, explosion doors lift	BMS low-fire permissive (10–20 % MAN)	S4 L1 4 Low
N9.4	Light-off LATE : Delayed ignition	• Igniter fault • poor atomisation	Fuel accumulates during the trial → puff	• 10-s trial for ignition then trip • purge before retry	S4 L2 8 Medium
N9.5	Sequence BEFORE : Fuel before feed flow	Step order not followed	Tubes fired dry → overheating	FSLL-008 as purge permissive and trip	S4 L1 4 Low
N9.7	Sequence AFTER : Feed stopped before the burner is out (shutdown)	Wrong order	Tubes overheated	FSLL-008 trip	S4 L1 4 Low

HAZOP recommendation

Start-up and shutdown procedures reviewed against this HAZOP; operators qualified on the simulator before first light-off.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
---------------	--	--	-----------------------------------	---

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			